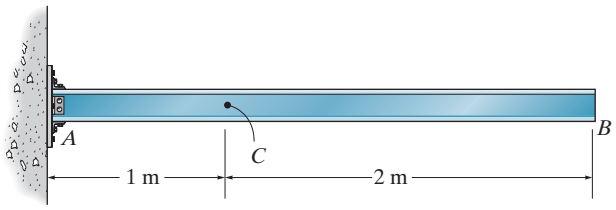
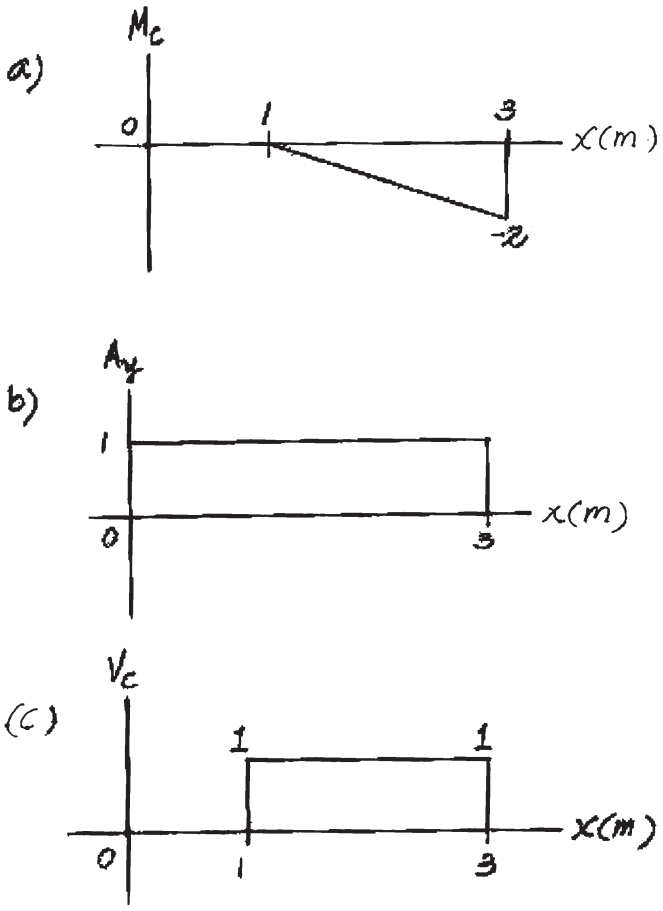
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**6–1.** Draw the influence lines for (a) the moment at $C$, (b) the vertical reaction at $A$, and (c) the shear at $C$. Assume $A$ is a fixed support. Solve this problem using the basic method of Sec. 6.1.



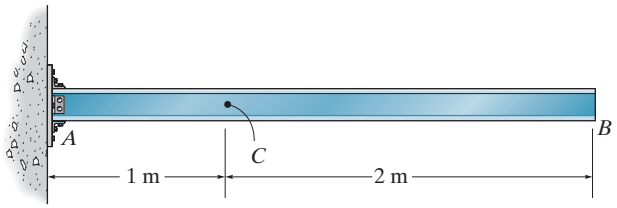

## SOLUTION




**Ans.**
$(M_C)_{\max} = -2$
$A_y = 1$
$(V_C)_{\max} = 1$

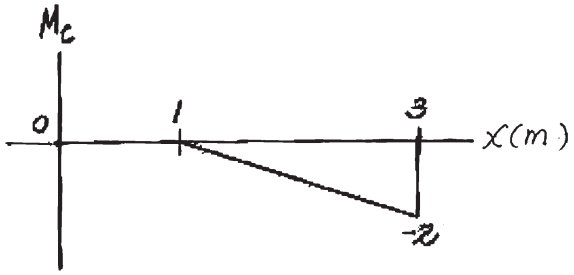
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**6–2.** Solve Prob. 6–1 using the Müller-Breslau principle.

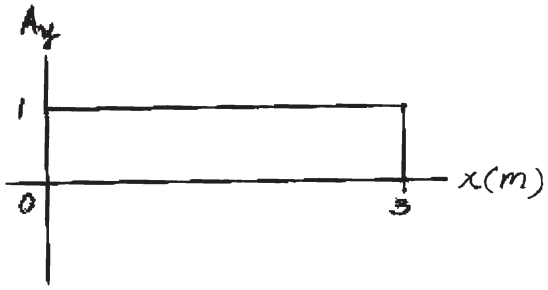



SOLUTION

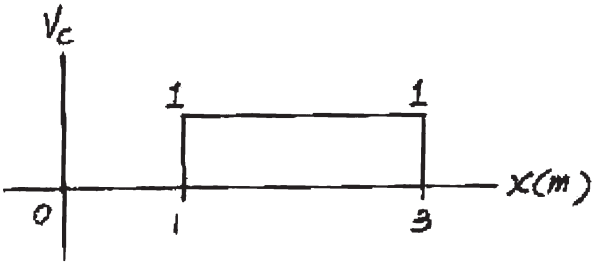
a)




b)

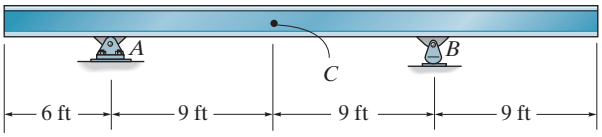



(c)



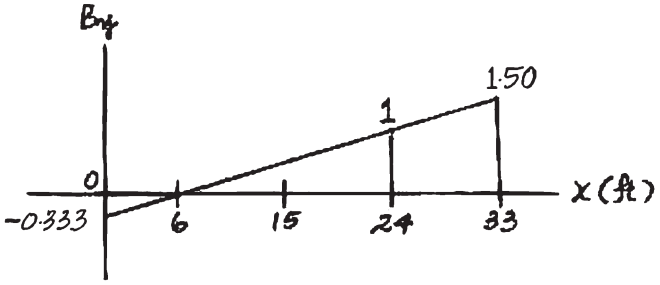
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**6–3.** Draw the influence lines for (a) the vertical reaction at $B$, (b) the moment at $C$, and (c) the shear just to the right of the support at $A$. Solve this problem using the basic method of Sec. 6.1.

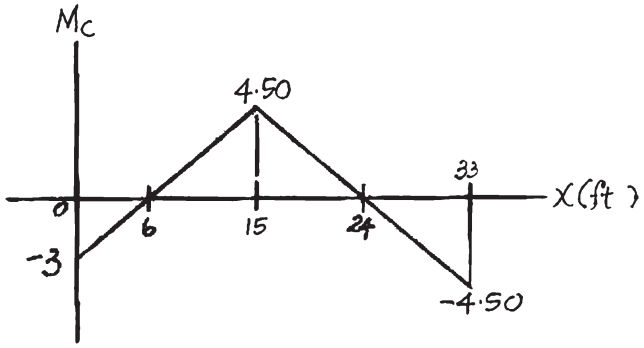



## SOLUTION

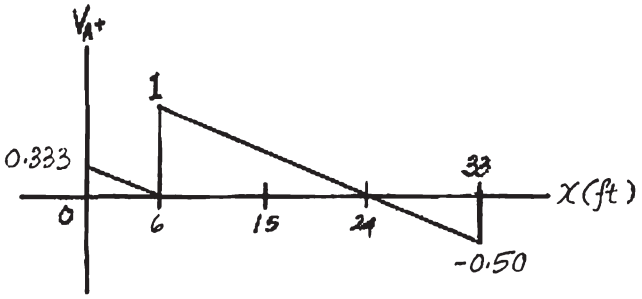
a)




b)



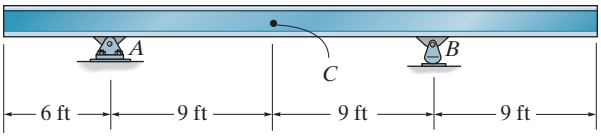

(c)




**Ans.**
$(V_B)_{max} = 1.5$
$(M_C)_{max} = \pm 4.5$
$(V_{A^+})_{max} = 1$

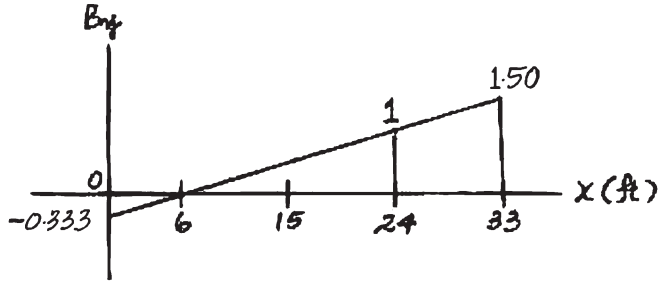
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***6–4.** Solve Prob. 6–3 using the Müller-Breslau principle.

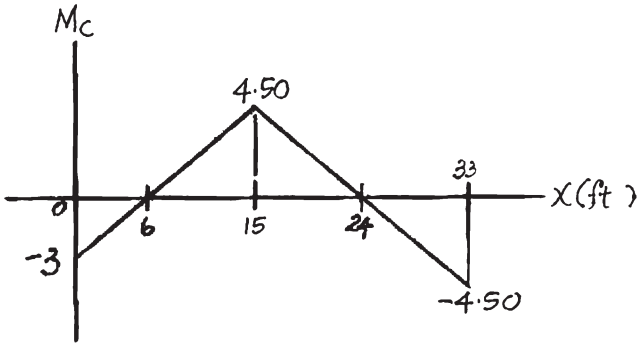



SOLUTION

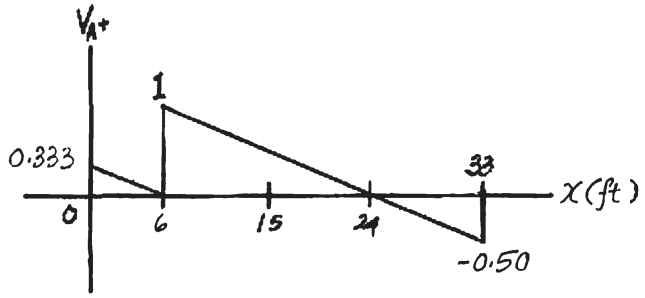
a)




b)

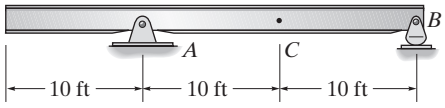



(c)



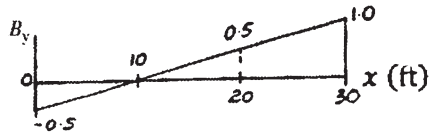
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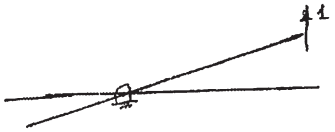
**6–5.** Draw the influence lines for (a) the vertical reaction at $B$, (b) the shear at $C$, and (c) the moment at $C$. Solve this problem using the basic method of Sec. 6–1.



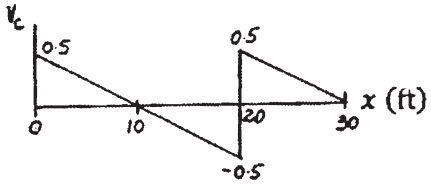

## SOLUTION

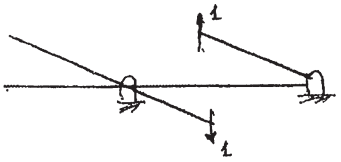
a)



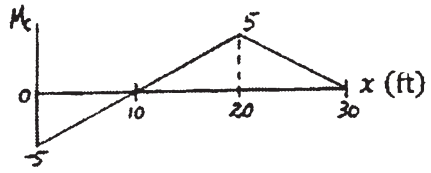



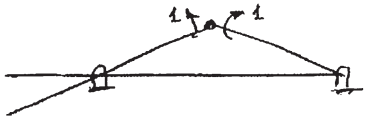

b)




c)

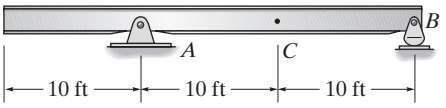



**Ans.**
$(B_y)_{max} = 1$
$(V_C)_{max} = \pm 0.5$
$(M_C)_{max} = \pm 5$

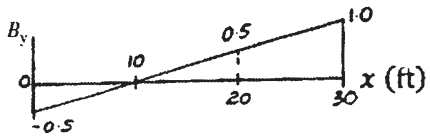
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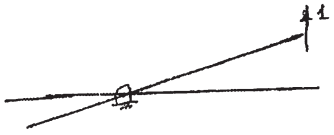
**6–6.** Solve Prob. 6–5 using the Müller-Breslau principle.



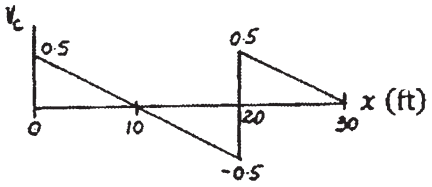

## SOLUTION

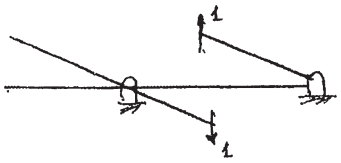
a)



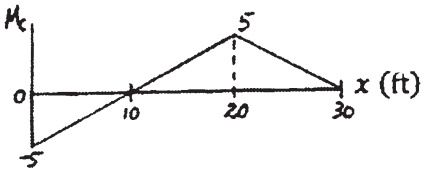



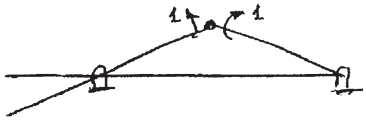

b)



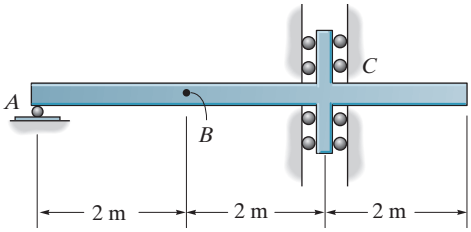

c)



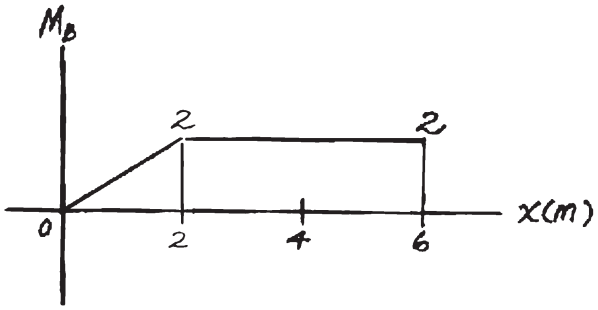
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**6–7.** Draw the influence lines for (a) the moment at $B$, (b) the shear at $B$, and (c) the vertical reaction at $A$. Solve this problem using the basic method of Sec. 6.1. *Hint*: The support at $C$ resists only a horizontal force and a bending moment.

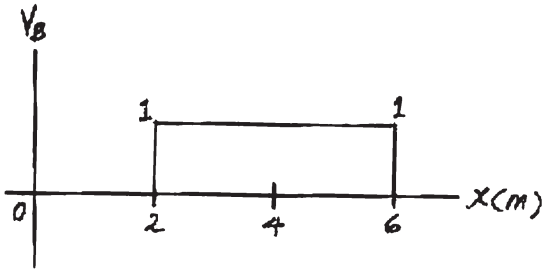



SOLUTION

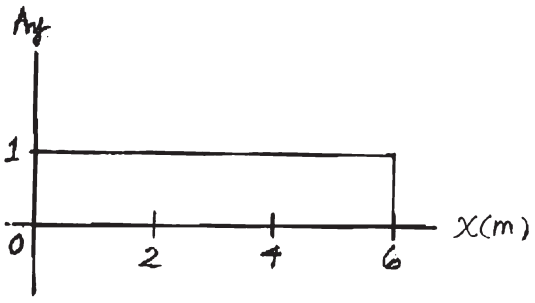
a)




b)



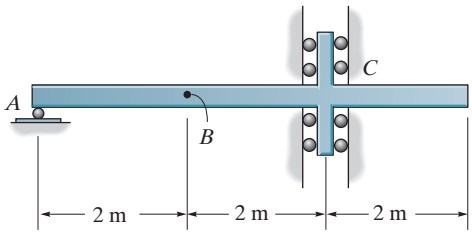

c)




**Ans.**
$(M_B)_{\max} = 2$
$(V_B)_{\max} = 1$
$A_y = 1$

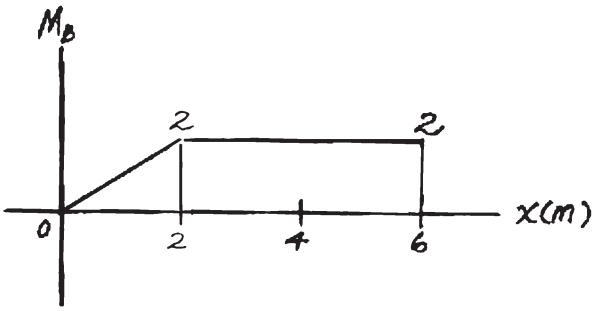
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***6–8.** Solve Prob. 6–7 using the Müller-Breslau principle.

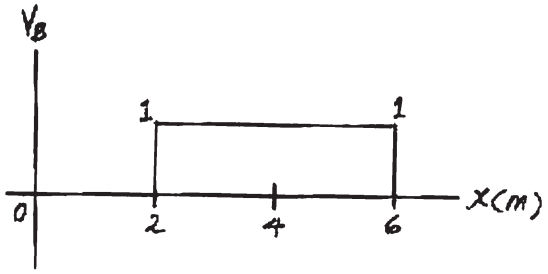



SOLUTION

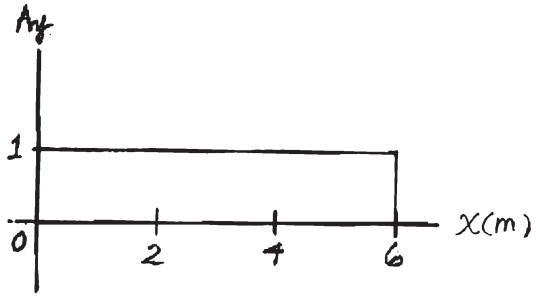
a)




b)

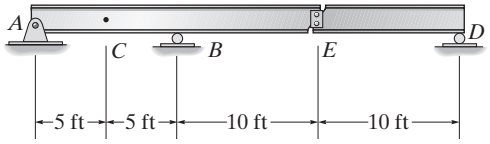



c)



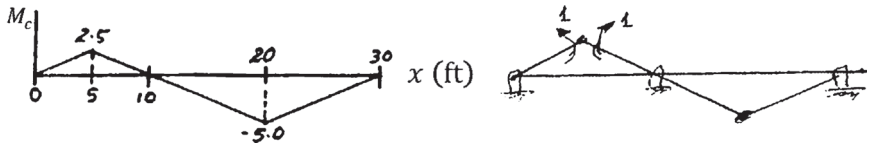
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**6–9.** Draw the influence lines for (a) the moment at $C$, (b) the vertical reaction at $A$, and (c) the vertical reaction at $B$. There is a short link at $E$. Solve this problem using the basic method of Sec. 6–1.

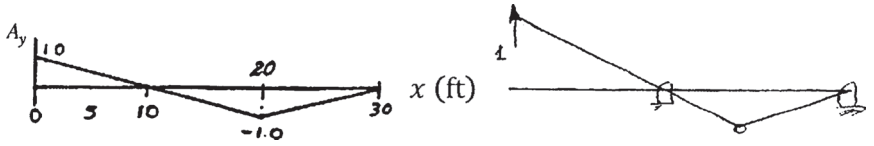



## SOLUTION

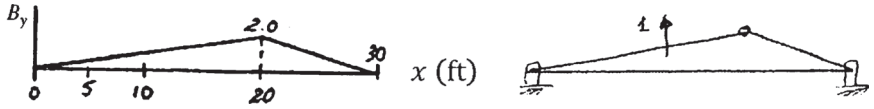
a)




b)



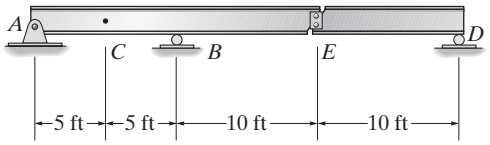

c)




**Ans.**
$(M_C)_{\max} = -5$
$(A_y)_{\max} = \pm 1$
$(B_y)_{\max} = 2$

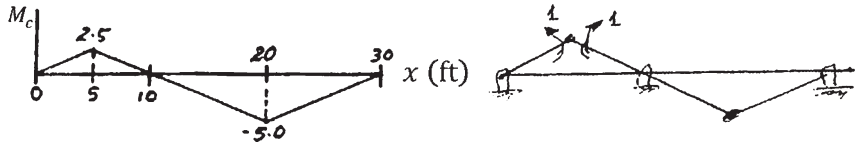
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**6–10.** Solve Prob. 6–9 using the Müller-Breslau principle.

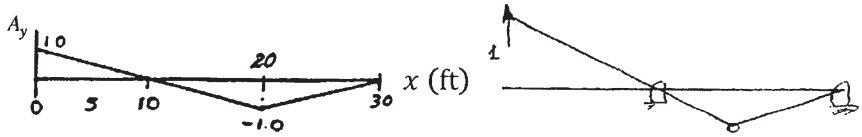



## SOLUTION

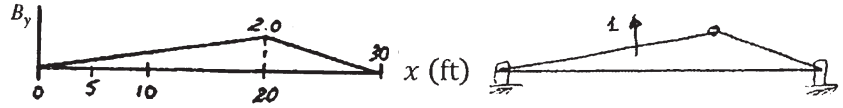
a)




b)

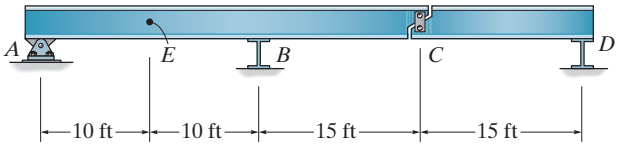



c)



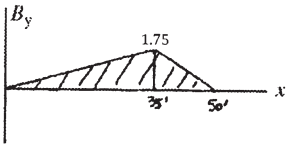
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**6–11.** Draw the influence lines for (a) the vertical reaction at $B$, and (b) the moment at $E$. Assume the supports at $B$ and $D$ are rollers. There is a short link at $C$. Solve this problem using the basic method of Sec. 6–1.




## SOLUTION

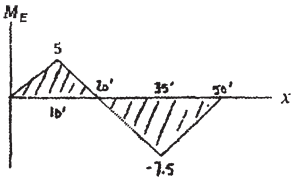
a)

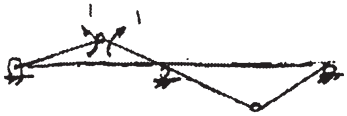






b)

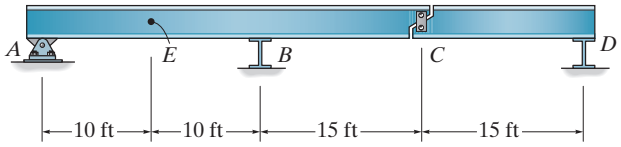



**Ans.**
$(B_y)_{\max} = 1.75$
$(M_E)_{\max} = -7.5$

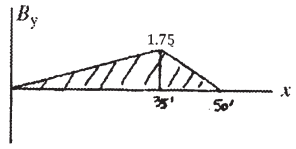
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***6–12.** Solve Prob. 6–11 using the Müller-Breslau principle.



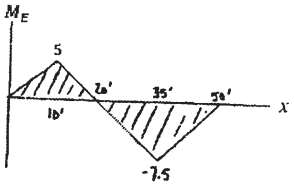

## SOLUTION

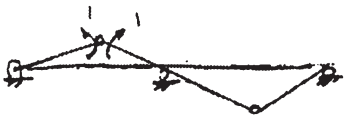
a)





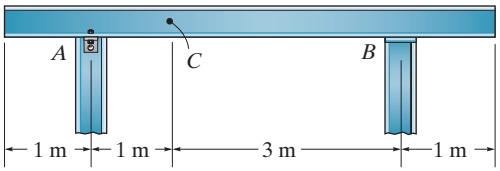

b)



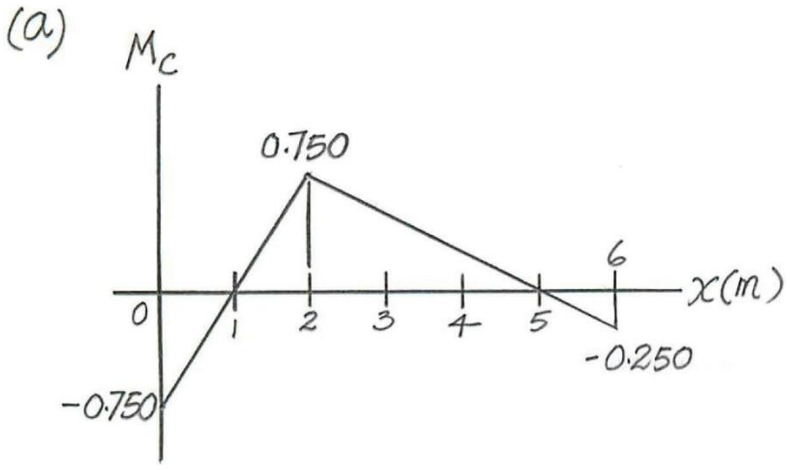



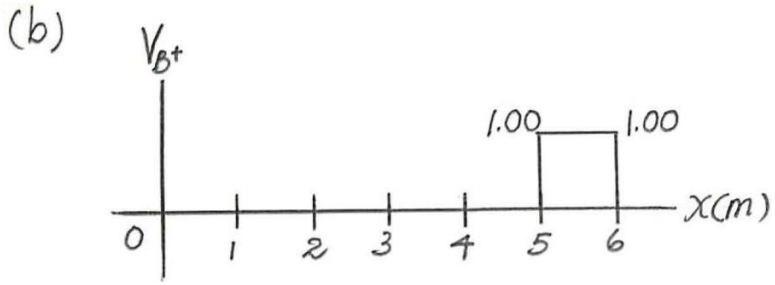
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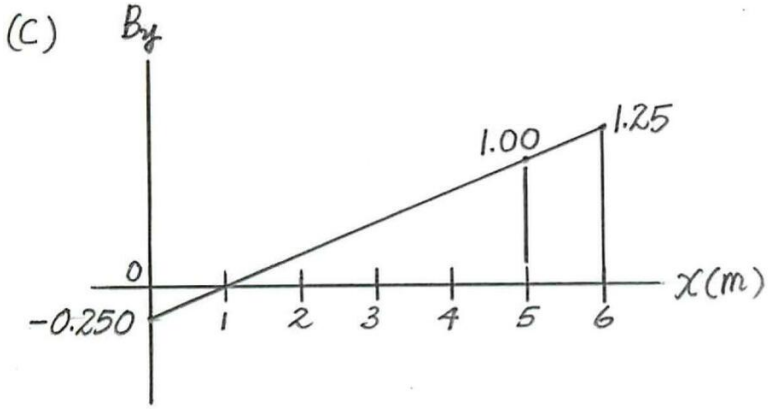
**6–13.** Draw the influence lines for (a) the moment at $C$, (b) the shear just to the right of the support at $B$, and (c) the vertical reaction at $B$. Solve this problem using the basic method of Sec. 6.1. Assume $A$ is a pin and $B$ is a roller.




## SOLUTION



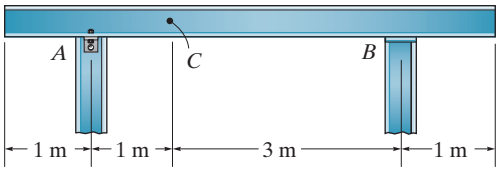




**Ans.**
$(M_C)_{\max} = \pm 0.75$
$(V_{B^+})_{\max} = 1$
$(B_y)_{\max} = 1.25$

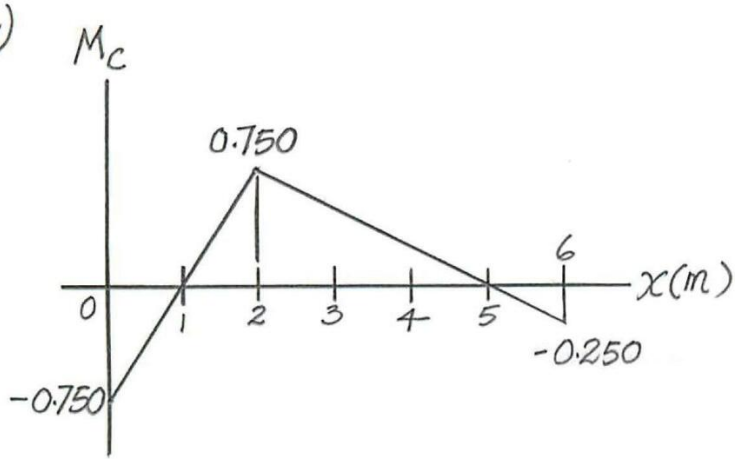
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**6–14.** Solve Prob. 6–13 using the Müller-Breslau principle.

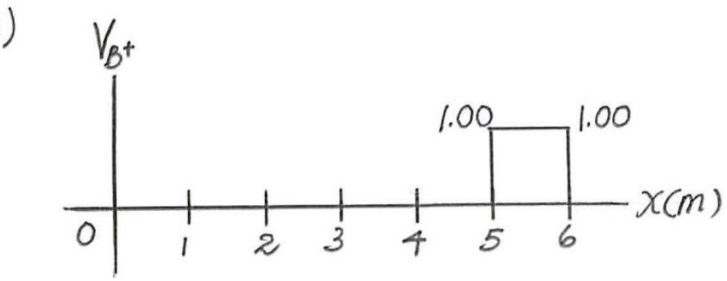



SOLUTION

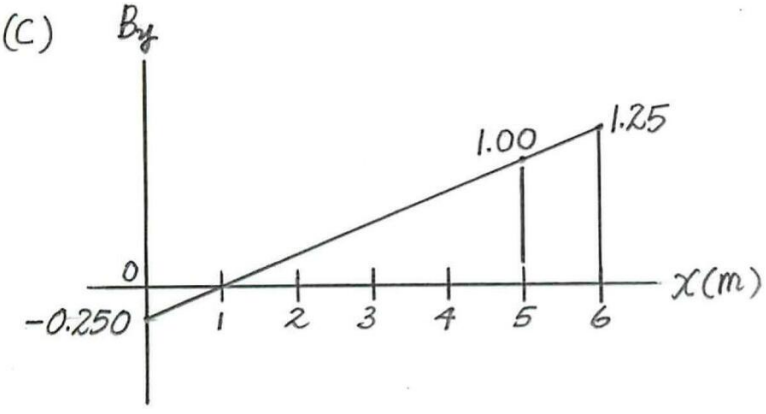
(a)




(b)

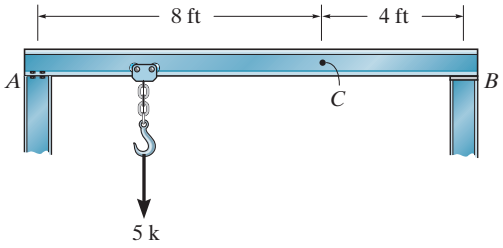



(c)

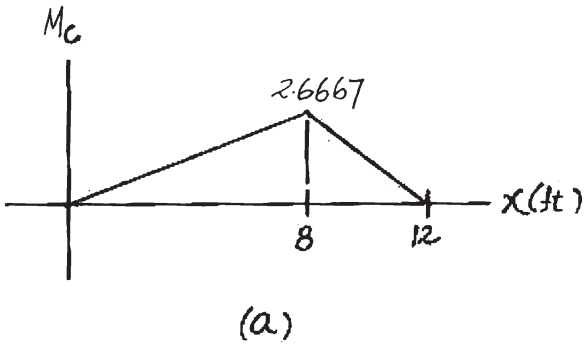


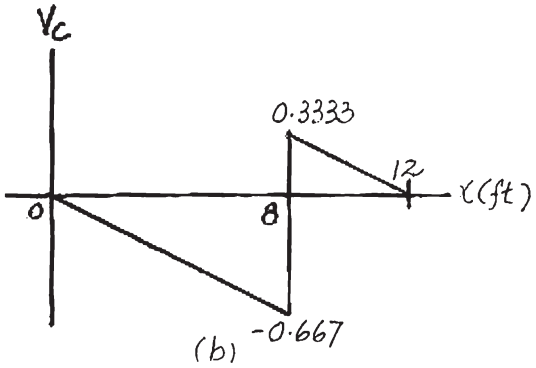
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**6–15.** The beam is subjected to a uniform dead load of 200 lb/ft and a single live load of 5 k. Determine (a) the maximum moment created by these loads at $C$, and (b) the maximum positive shear at $C$. Assume $A$ is a pin, and $B$ is a roller.




## SOLUTION




Referring to the influence line for the moment at $C$ shown in Fig. *a,* the maximum moment at $C$ is

$$(M_C)_{\max} = 2.6667(5) + \frac{1}{2}(12)(2.6667)(0.2)$$

$$= 16.53\ \text{k}\cdot\text{ft} = 16.5\ \text{k}\cdot\text{ft} \qquad \textbf{Ans.}$$

Referring to the influence line for shear at $C$ shown in Fig. *b,* the maximum positive shear at $C$ is

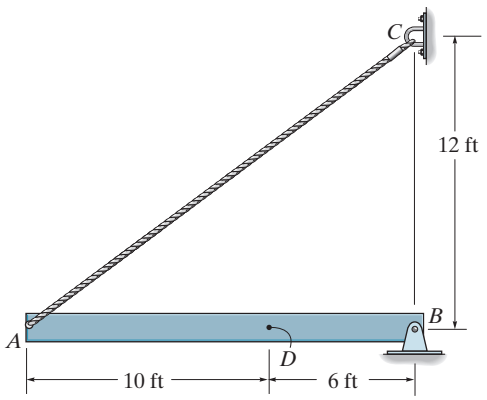
$$(V_C)_{\max} = \left[\frac{1}{2}(8)(-0.6667)\right](0.2) + \left[\frac{1}{2}(12.8)(0.3333)\right](0.2) + 5(0.333)$$

$$= 1.2667\ \text{k} = 1.27\ \text{k} \qquad \textbf{Ans.}$$

**Ans.**
$(M_C)_{\max} = 16.5\ \text{k}\cdot\text{ft};$
$(V_C)_{\max} = 1.27\ \text{k}$

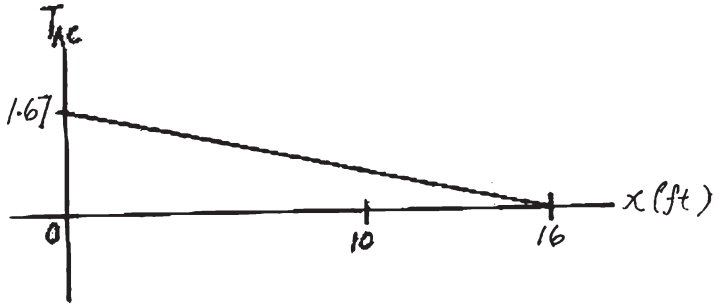
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***6–16.** Draw the influence lines for (a) the force in the cable *AC*, (b) the vertical reaction at *B*, and (c) the moment at *D*.

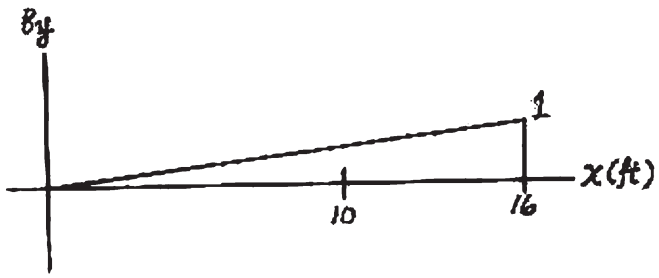



SOLUTION

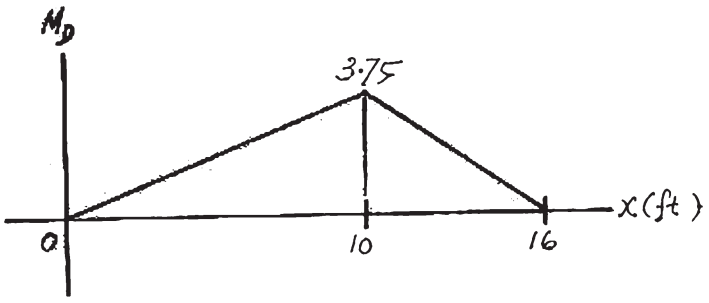
(a)




(b)

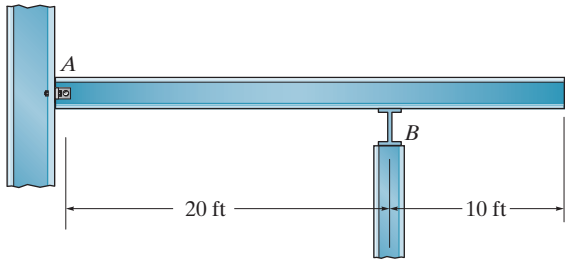



(c)



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**6–17.** A uniform live load of 300 lb/ft and a single live concentrated force of 1500 lb are to be placed on the beam. The beam has a weight of 150 lb/ft. Determine (a) the maximum vertical reaction at support $B$, and (b) the maximum negative moment at $B$. Assume the support at $A$ is a pin and $B$ is a roller.




## SOLUTION

Referring to the influence line for the vertical reaction at $B$ shown in Fig. *a,* the maximum reaction is

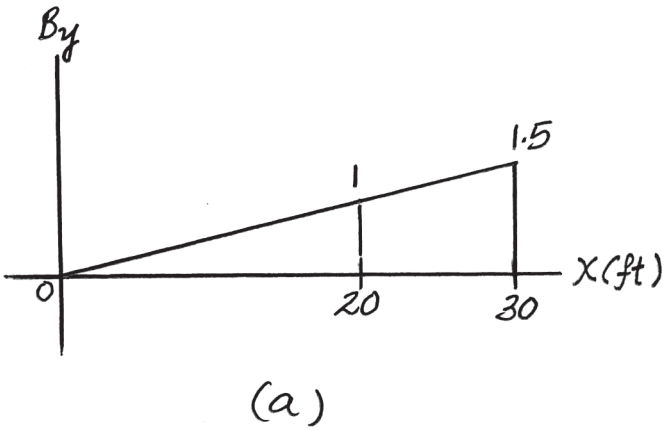
$$(B_y)_{\max(+)} = 1.5(1500) + \left[\frac{1}{2}(30 - 0)(1.5)\right](300 + 150)$$

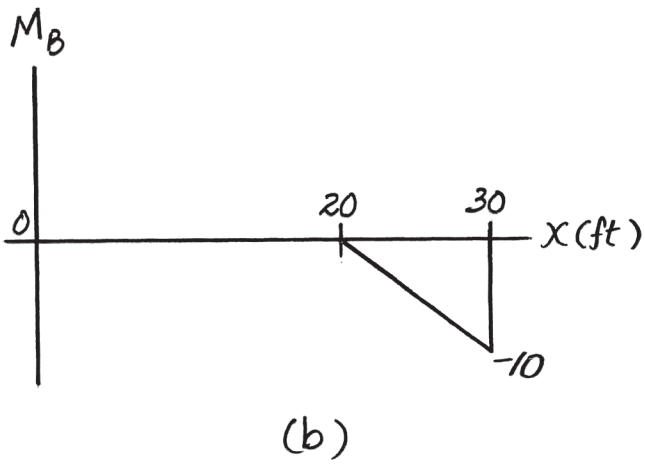
$$= 12375\ \text{lb} = 12.4\ \text{k} \qquad \textbf{Ans.}$$

Referring to the influence line for the moment at $B$ shown in Fig. *b,* the maximum negative moment is

$$(M_B)_{\max(-)} = -10(1500) + \left[\frac{1}{2}(30 - 20)(-10)\right](300 + 150)$$

$$= -37500\ \text{lb}\cdot\text{ft} = -37.5\ \text{k}\cdot\text{ft} \qquad \textbf{Ans.}$$

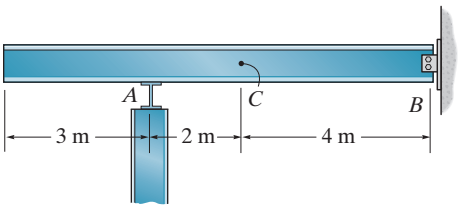


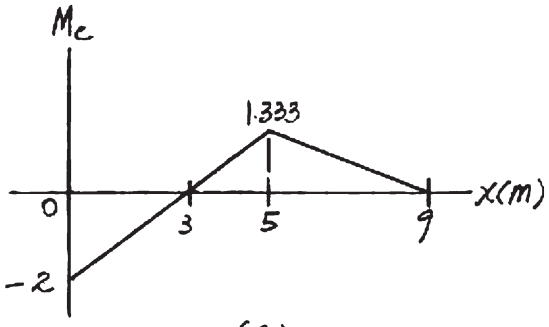

**Ans.**
$(B_y)_{\max(+)} = 12.4\ \text{k}$;
$(M_B)_{\max(-)} = -37.5\ \text{k}\cdot\text{ft}$

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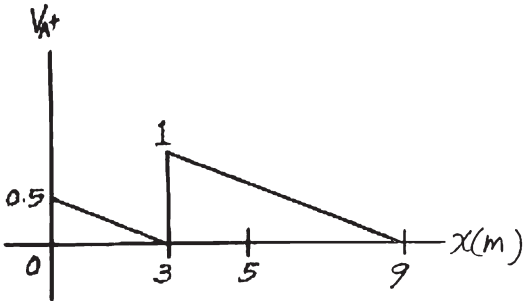
**6–18.** The beam supports a uniform dead load of 6 kN/m, a live load of 20 kN/m, and a single live concentrated force of 40 kN. Determine (a) the maximum positive moment at $C$, and (b) the maximum positive shear just to the right of the support at $A$. Assume $A$ is a roller and $B$ is a pin.




## SOLUTION




(a)




(b)

Referring to the influence line for the moment at $C$ shown in Fig. *a,* the maximum positive moment at $C$ is

$$(M_C)_{\max(+)} = (40)(1.3333) + \left[\frac{1}{2}(3)(-2)\right](6) + \left[\frac{1}{2}(9-3)(1.333)\right](6) + \left[\frac{1}{2}(9-3)(1.333)\right](20)$$
$$= 139.33\ \text{kN}\cdot\text{m} = 139\ \text{kN}\cdot\text{m} \qquad \textbf{Ans.}$$

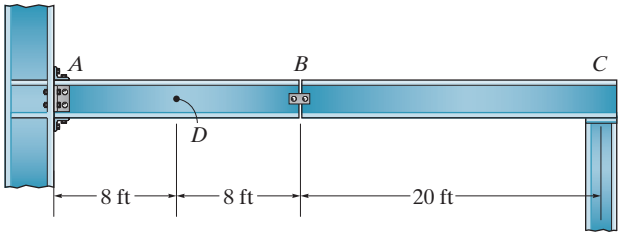
Referring to the influence line for the shear at a point just to the right of support $A$ shown in Fig. *b*, the maximum positive shear at this point is

$$(V_{A^+})_{\max(+)} = \left[\frac{1}{2}(3)(0.5)\right](6) + \left[\frac{1}{2}(9-3)(1)\right](6) + \left[\frac{1}{2}(3)(0.5)\right](20) + \left[\frac{1}{2}(9-3)(1)\right](20) + 40(1)$$
$$= 137.5\ \text{kN} \qquad \textbf{Ans.}$$

**Ans.**
$(M_C)_{\max(+)} = 139\ \text{kN}\cdot\text{m};$
$(V_{A^+})_{\max(+)} = 137.5\ \text{kN}$

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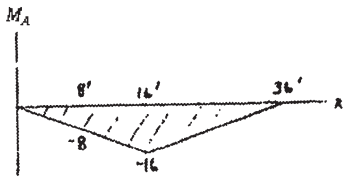
**6–19.** Where should the beam $ABC$ be loaded with a 300-lb/ft uniform distributed live load so it causes (a) the largest live moment at point $A$ and (b) the largest live shear at $D$? Calculate the values of the moment and shear. Assume the support at $A$ is fixed, $B$ is pinned and $C$ is a roller.

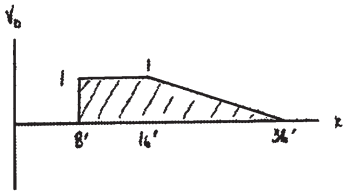


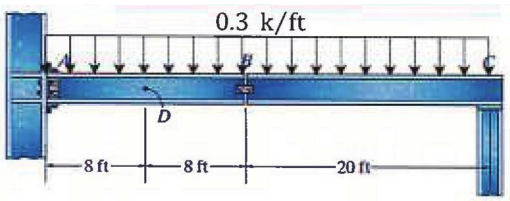

## SOLUTION

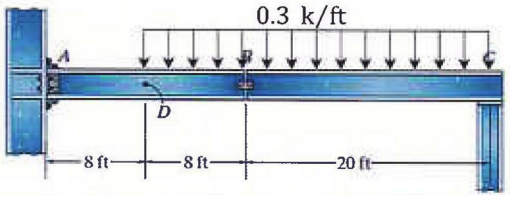
(a) $(M_A)_{\max} = \frac{1}{2}(36)(-16)(0.3) = -86.4\ \text{k}\cdot\text{ft}$ **Ans.**

(b) $(V_D)_{\max} = \left[(1)(8) + \frac{1}{2}(1)(20)\right](0.3) = 5.40\ \text{k}$ **Ans.**



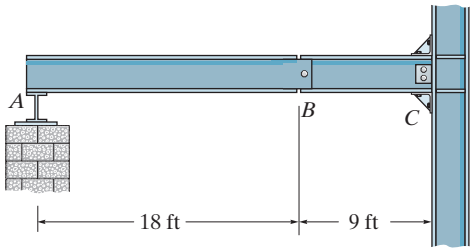




**Ans.**
$(M_A)_{\max} = -86.4\ \text{k}\cdot\text{ft}$;
$(V_D)_{\max} = 5.40\ \text{k}$

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***6–20.** The compound beam is subjected to a uniform dead load of 200 lb/ft and a single live load of 2 k. Determine (a) the maximum negative moment created by these loads at $C$, and (b) the maximum negative shear at $B$. Assume $C$ is a fixed support, $B$ is a pin, and $A$ is a roller.



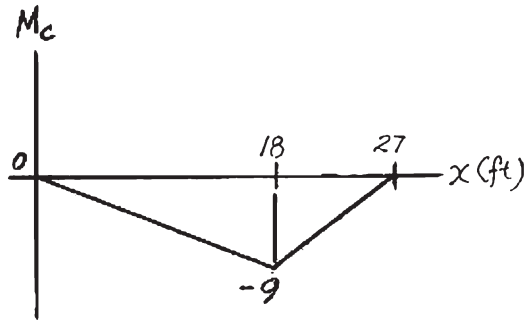

## SOLUTION

Referring to the influence line for the moment at $C$ shown in Fig. $a$, the maximum negative moment at $C$ is

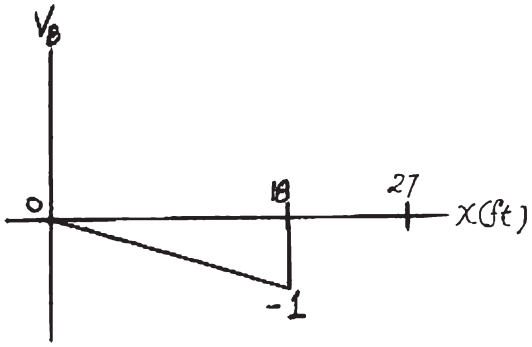
$$(M_C)_{\max(-)} = \left[\frac{1}{2}(27)(-9)\right](0.2) + 2(-9) = -42.3\ \text{k}\cdot\text{ft} \qquad \textbf{Ans.}$$

Referring to the influence line for the shear at $B$ shown in Fig. $b$, the maximum negative shear at $B$ is

$$(V_B)_{\max(-)} = \left[\frac{1}{2}(18)(-1)\right](0.2) + 2(-1) = -3.80\ \text{k} \qquad \textbf{Ans.}$$




(a)

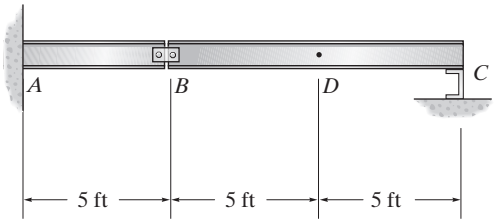



(b)

**Ans.**
$(M_C)_{\max(-)} = -42.3\ \text{k}\cdot\text{ft};$
$(V_B)_{\max(-)} = -3.80\ \text{k}$

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**6–21.** The compound beam is subjected to a uniform dead load of 200 lb/ft and a uniform live load of 150 lb/ft. Determine (a) the maximum negative moment these loads develop at $A$, and (b) the maximum positive shear at $D$. Assume $B$ is a pin and $C$ is a roller.



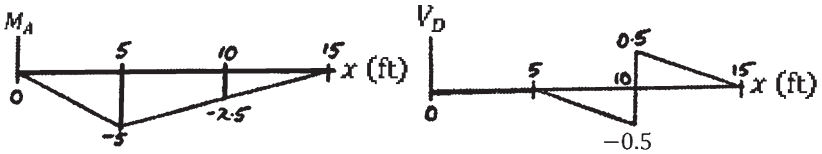

## SOLUTION

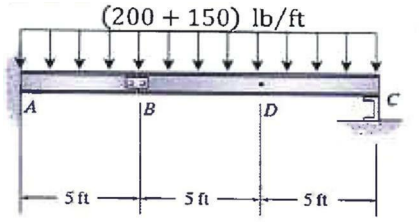
**a)** $(M_A)_{\max(-)} = (200 + 150)\left(\frac{1}{2}\right)(-5)(15)$

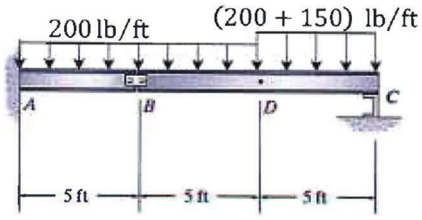
$= -13125\ \text{lb}\cdot\text{ft}$ **Ans.**

**b)** $(V_D)_{\max(+)} = (200)\left(\frac{1}{2}\right)(5)(-0.5) + (200 + 150)\left(\frac{1}{2}\right)(5)(0.5)$

$= 188\ \text{lb}$ **Ans.**



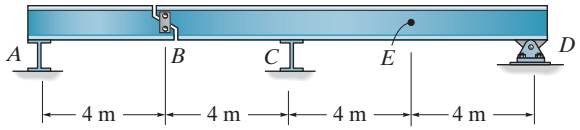




**Ans.**
$(M_A)_{\max(-)} = -13{,}125\ \text{lb}\cdot\text{ft}$;
$(V_D)_{\max(+)} = 188\ \text{lb}$

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**6–22.** The beam is subjected to a uniform live load of 1.2 kN/m, a dead load of 0.5 kN/m, and a single live load of 40 kN. Determine (a) the maximum positive moment created by these loads at $E$, and (b) the maximum positive shear at $E$. Assume $A$ and $C$ are rollers, and $B$ is a short link.



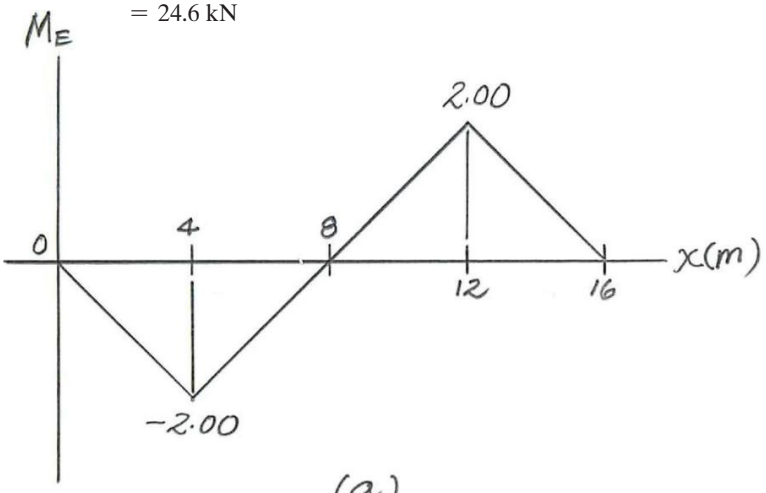

## SOLUTION

Referring to the influence line for the moment at $E$, shown in Fig. $a$, the maximum positive moment at this point is

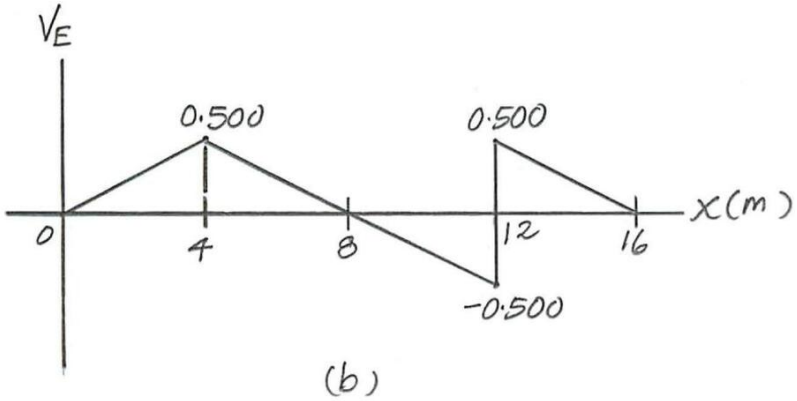
$$(M_E)_{\max(+)} = 40(2.00) + \left[\frac{1}{2}(16 - 8)(2.00)\right](1.2 + 0.5) + \left[\frac{1}{2}(8 - 0)(-2.00)\right](0.5)$$
$$= 89.6\ \text{kN}\cdot\text{m} \qquad \textbf{Ans.}$$

Referring to the influence line for the shear at $E$, shown in Fig. $b$, the maximum positive shear at this point is

$$(V_E)_{\max(+)} = 40(0.5) + \left[\frac{1}{2}(8 - 0)(0.5)\right](1.2 + 0.5) + \left[\frac{1}{2}(16 - 12)(0.5)\right](1.2 + 0.5)$$
$$+ \left[\frac{1}{2}(12 - 8)(-0.5)\right](0.5)$$
$$= 24.6\ \text{kN} \qquad \textbf{Ans.}$$




(a)

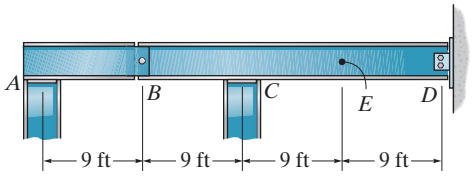



(b)

**Ans.**
$(M_E)_{\max(+)} = 89.6\ \text{kN}\cdot\text{m}$;
$(V_E)_{\max(+)} = 24.6\ \text{kN}$

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**6–23.** The beam supports a uniform dead load of 500 lb/ft and a single live concentrated force of 5 k. Determine (a) the maximum negative moment at $E$, and (b) the maximum positive shear at $E$. Assume the support at $D$ is a pin, $A$ and $C$ are rollers, and $B$ is a pin.



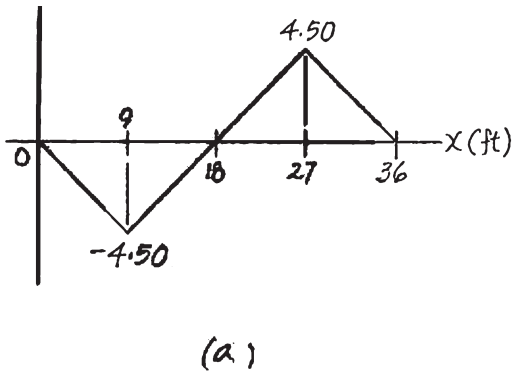

## SOLUTION

Referring to the influence line for the moment at $E$ shown in Fig. *a,* the maximum negative moment at $E$ is

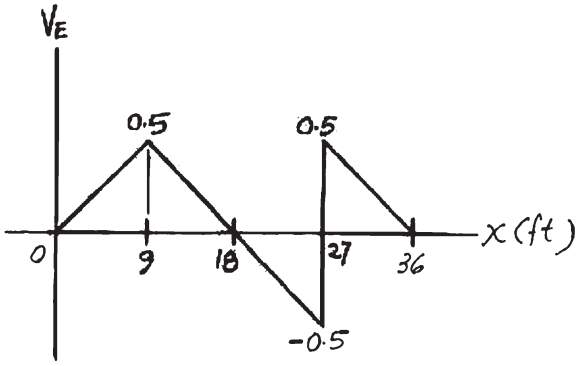
$$(M_E)_{\max(-)} = \left[\frac{1}{2}(18)(-4.50)\right](0.5) + \left[\frac{1}{2}(36 - 18)(4.50)\right](0.5) + 5(-4.50)$$
$$= -22.5\ \text{k}\cdot\text{ft} \qquad \textbf{Ans.}$$

Referring to the influence line for the shear at $E$ shown in Fig. *b,* the maximum positive shear at $E$ is

$$(V_E)_{\max(+)} = \left[\frac{1}{2}(18)(0.5)\right](0.5) + \left[\frac{1}{2}(27 - 18)(0.5)\right](0.5) + \left[\frac{1}{2}(36 - 27)(0.5)\right](0.5) + 5(0.5)$$
$$= 4.75\ \text{k} \qquad \textbf{Ans.}$$




(a)

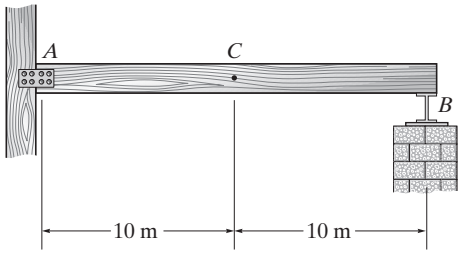



(b)

**Ans.**
$(M_E)_{\max(-)} = -22.5\ \text{k}\cdot\text{ft};$
$(V_E)_{\max(+)} = 4.75\ \text{k}$

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***6–24.** The beam supports a uniform dead load of 500 N/m and single live concentrated force of 3000 N. Determine (a) the maximum positive moment that can be developed at point *C*, and (b) the maximum positive shear that can be developed at point *C*. Assume the support at *A* is a pin and *B* is a roller.




## SOLUTION

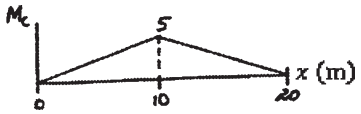
**a)** $(M_C)_{\max} = 500\left(\frac{1}{2}\right)(5)(20) + 3000(5)$

$= 40\,000\ \text{N}\cdot\text{m} = 40.0\ \text{kN}\cdot\text{m}$ **Ans.**

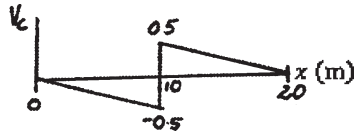
**b)** $(V_C)_{\max} = 500\left(\frac{1}{2}\right)(0.5)(10) + 3000(0.5) - 500\left(\frac{1}{2}\right)(0.5)(10)$

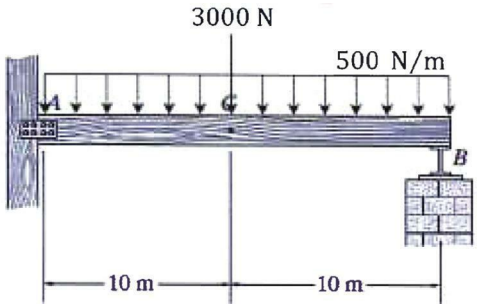
$= 1500\ \text{N} = 1.50\ \text{kN}$ **Ans.**

a)




b)

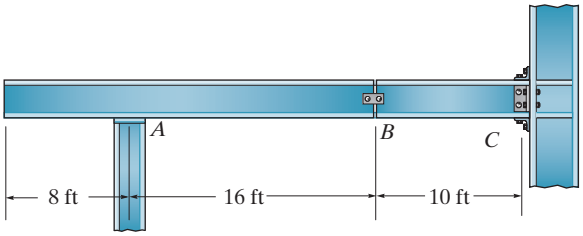



**Ans.**
$(M_C)_{\max} = 40.0\ \text{kN}\cdot\text{m}$;
$(V_C)_{\max} = 1.50\ \text{kN}$

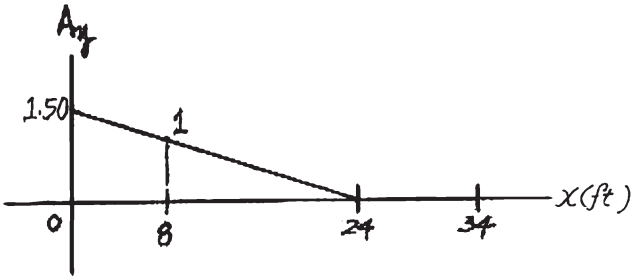
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**6–25.** Draw the influence lines for (a) the vertical reaction at $A$, (b) the shear just to the right of $A$, and (c) the moment at $C$. Assume $C$ is fixed, $A$ is a roller, and $B$ is a pin.

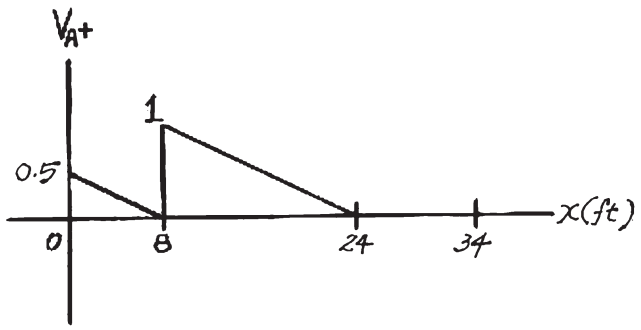



## SOLUTION

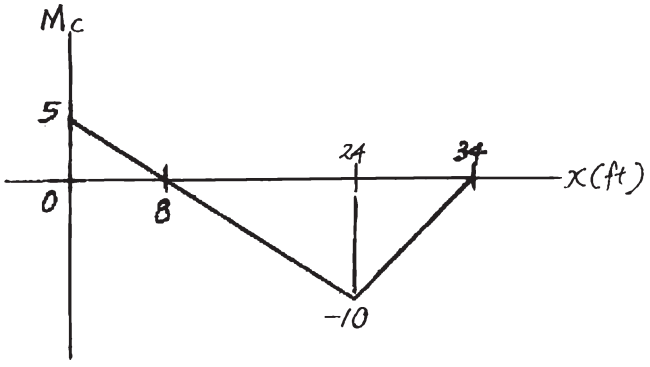
a)




b)



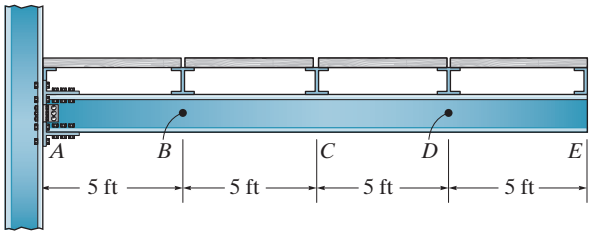

c)




**Ans.**
$(A_y)_{max} = 1.5$
$(V_{A^+})_{max} = 1$
$(M_C)_{max} = -10$

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**6–26.** A uniform live load of 0.4 k/ft and a concentrated live force of 2 k are to be placed on the floor slabs. Determine (a) the maximum live shear in panel *CD*, and (b) the maximum live moment at *B*.



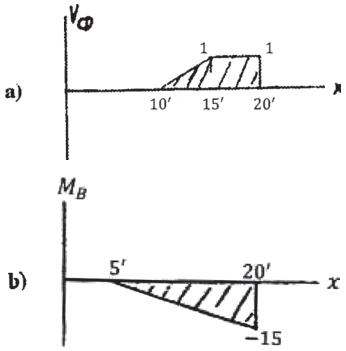

## SOLUTION

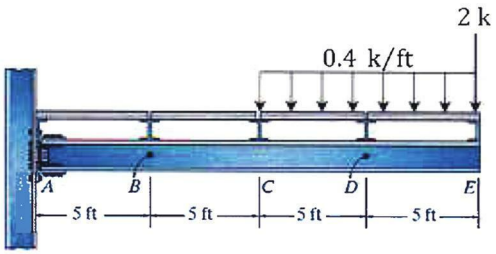
**a)** $(V_{CD})_{\max(+)} = 2(1) + 0.4\left[\frac{1}{2}(1)(5) + (1)(5)\right]$

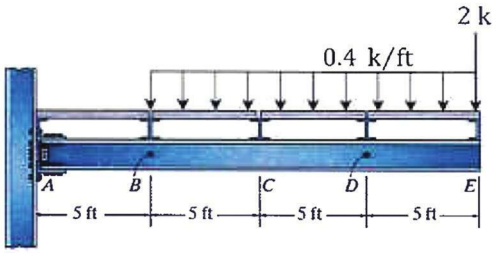
$= 5\text{ k}$ **Ans.**

**b)** $(M_B)_{\max(-)} = 2(-15) + 0.4\left(\frac{1}{2}\right)(-15)(15)$

$= -75\text{ k}\cdot\text{ft}$ **Ans.**



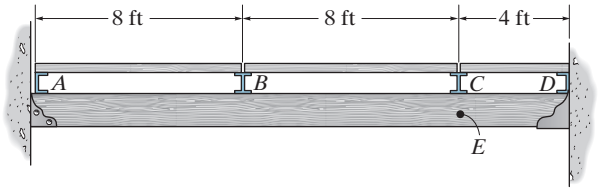




**Ans.**
$(V_{CD})_{\max(+)} = 5\text{ k}$;
$(M_B)_{\max(-)} = -75\text{ k}\cdot\text{ft}$

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**6–27.** Draw the influence line for the moment at $E$ in the girder. Determine the maximum positive moment in the girder at $E$ if a single concentrated live force of 1200 lb and a uniform live load of 200 lb/ft can be placed on the floor beams. Assume $A$ is a pin and $D$ is a roller.

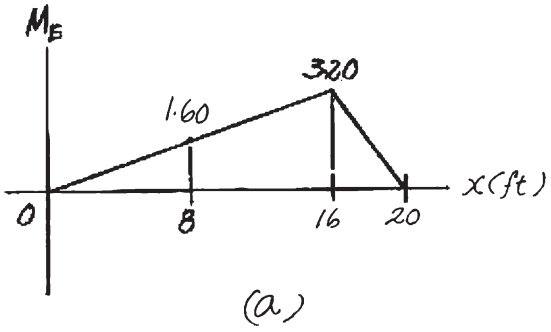



## SOLUTION

Referring to the influence line for the moment at point $E$ shown in Fig. *a*,

$$(M_E)_{\max(+)} = \left[\frac{1}{2}(20)(3.20)\right](200) + 1200(3.20)$$

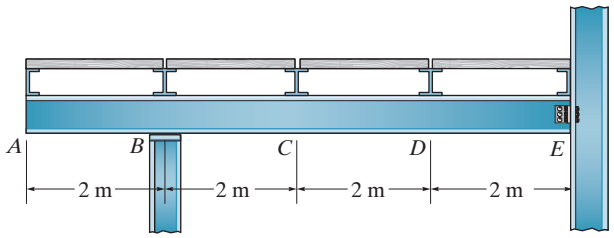
$$= 10.24(10^3)\ \text{lb}\cdot\text{ft} = 10.2\ \text{k}\cdot\text{ft} \qquad \textbf{Ans.}$$




**Ans.**
$(M_E)_{\max(+)} = 10.2\ \text{k}\cdot\text{ft}$

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***6–28.** A uniform live load of 20 kN/m and a single concentrated live force of 30 kN are placed on the floor beams. If the beams also support a uniform dead load of 4 kN/m, determine (a) the maximum positive shear in panel *CD* of the girder and (b) the maximum negative moment in the girder at *C*. Assume the support at *B* is a roller and *E* is a pin.



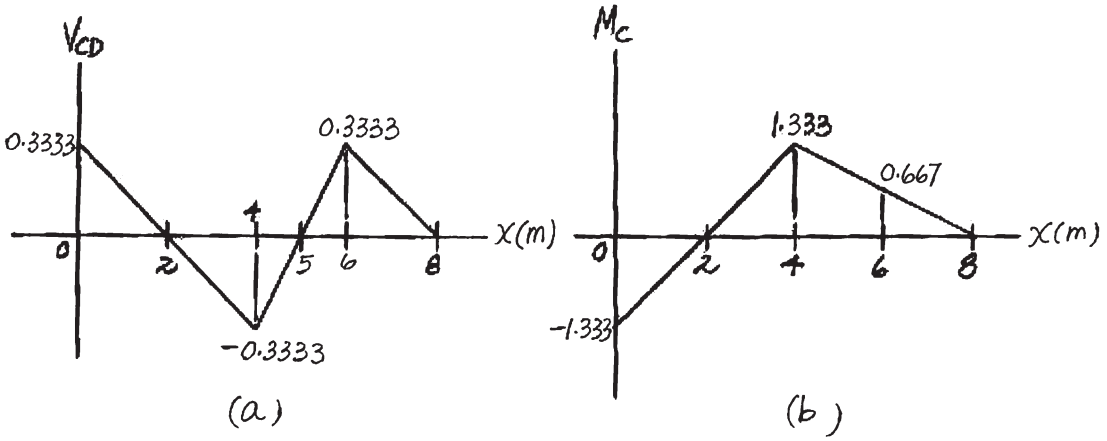

## SOLUTION

Referring to the influence line for the shear in panel *CD* shown in Fig. *a*, the maximum positive shear within this panel is

$$(V_{CD})_{\max(+)} = \left[\frac{1}{2}(2)(0.3333)\right](20 + 4) + \left[\frac{1}{2}(8 - 5)(0.3333)\right](20 + 4)$$
$$+ \left[\frac{1}{2}(5 - 2)(-0.3333)\right](4) + 30(0.3333)$$
$$= 28.0\ \text{kN} \qquad \textbf{Ans.}$$

Referring to the influence line for the moment at *C* shown in Fig. *b*, the maximum negative moment at this point is

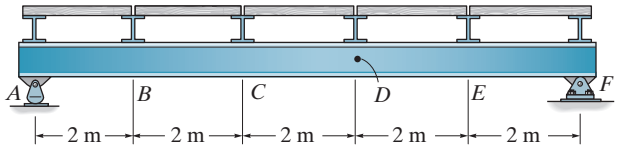
$$(M_C)_{\max(-)} = \left[\frac{1}{2}(2)(-1.333)\right](20 + 4) + \left[\frac{1}{2}(8 - 2)(1.3333)\right](4) + 30(-1.333)$$
$$= -56.0\ \text{kN}\cdot\text{m} \qquad \textbf{Ans.}$$



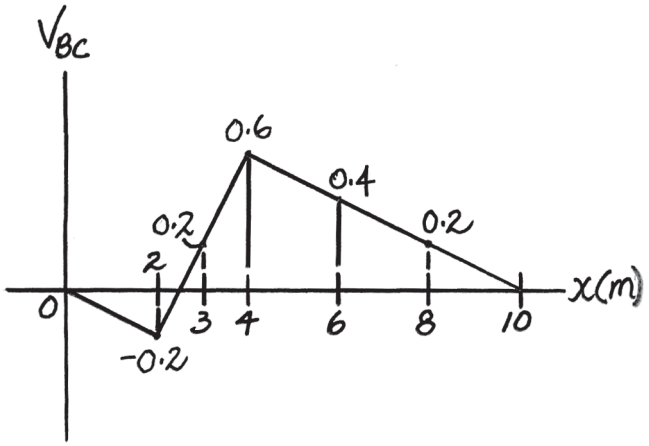

**Ans.**
$(V_{CD})_{\max(+)} = 28.0\ \text{kN}$;
$(M_C)_{\max(-)} = -56.0\ \text{kN}\cdot\text{m}$

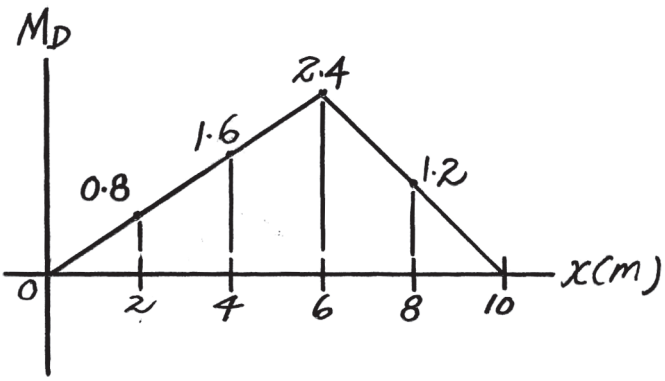
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**6–29.** Draw the influence lines for (a) the shear in panel *BC* of the girder, and (b) the moment at *D*.




SOLUTION

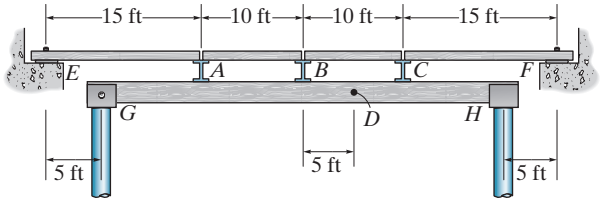



**Ans.**
$(V_{BC})_{max} = 0.6$
$(M_D)_{max} = 2.4$

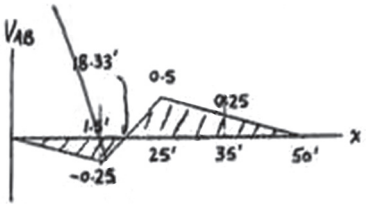
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**6–30.** A uniform live load of 250 lb/ft and a single concentrated live force of 1.5 k are to be placed on the floor beams. Determine (a) the maximum positive shear in panel $AB$, and (b) the maximum moment at $D$. Assume only vertical reactions occur at the supports.

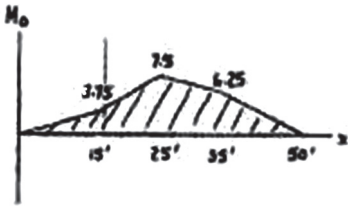



## SOLUTION

$$(V_{AB})_{\max} = \frac{1}{2}(25 - 18.33)(0.5)(0.250) + \frac{1}{2}(0.5 + 0.25)(35 - 25)(0.250)$$
$$+ \frac{1}{2}(50 - 35)(0.25)(0.250) + 0.5(1.5) = 2.57\ \text{k} \qquad \textbf{Ans.}$$



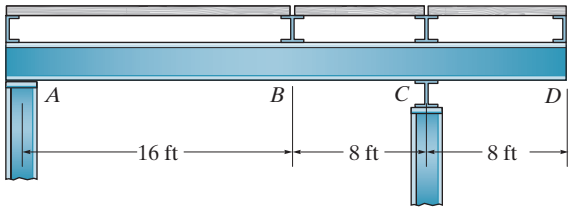

$$(M_D)_{\max} = \left[\frac{1}{2}(3.75)(15) + \frac{1}{2}(3.75 + 7.5)(10) + \frac{1}{2}(7.5 + 6.25)(10)\right.$$
$$\left. + \frac{1}{2}(6.25)(15)\right](0.250) + 7.5(1.5)$$
$$= 61.25\ \text{k} \cdot \text{ft} \qquad \textbf{Ans.}$$




**Ans.**
$(V_{AB})_{\max} = 2.57\ \text{k};$
$(M_D)_{\max} = 61.25\ \text{k} \cdot \text{ft}$

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**6–31.** A uniform live load of 200 lb/ft and a single concentrated live force of 1500 lb are placed on the top beams. If the beams also support a uniform dead load of 50 lb/ft, determine (a) the maximum negative shear in panel *BC* of the girder and (b) the maximum positive moment in the girder at *B*. Assume *C* is a roller and *A* is a pin.




## SOLUTION

Referring to the influence line for the shear within panel *BC* shown in Fig. *a*, the maximum negative shear within this panel is

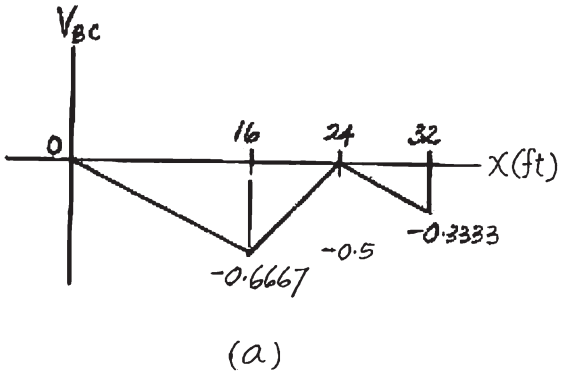
$$(V_{BC})_{\max(-)} = \left[\frac{1}{2}(24)(-0.6667)\right](0.2 + 0.05) + \left[\frac{1}{2}(-0.3333)(32 - 24)\right](0.2 + 0.05) + 1.5(-0.6667)$$

$$= -3.333\text{ k} = -3.33\text{ k} \quad \textbf{Ans.}$$

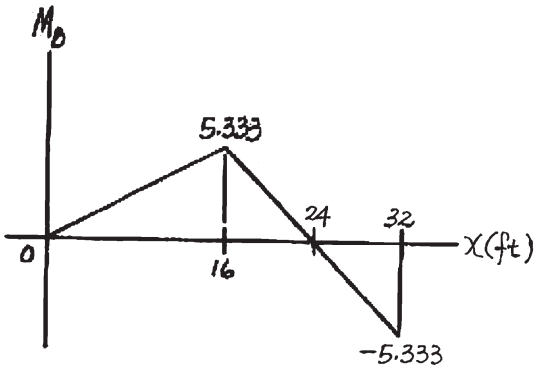
Referring to the influence line for the moment at *B* shown in Fig. *b*, the maximum positive moment at *B* is

$$(M_B)_{\max(+)} = \left[\frac{1}{2}(24)(5.333)\right](0.2 + 0.05) + \left[\frac{1}{2}(32 - 24)(-5.333)\right](0.05) + 1.5(5.333)$$

$$= 22.93\text{ k}\cdot\text{ft} = 22.9\text{ k}\cdot\text{ft} \quad \textbf{Ans.}$$




(a)

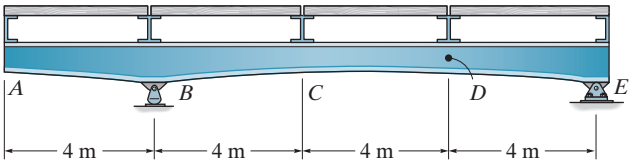



(b)

**Ans.**
$(V_{BC})_{\max(-)} = -3.33\text{ k};$
$(M_B)_{\max(+)} = 22.9\text{ k}\cdot\text{ft}$

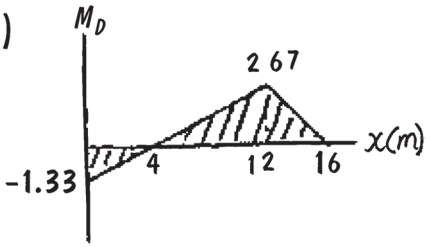
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***6–32.** Draw the influence lines for (a) the moment at *D* in the girder, and (b) the shear in panel *BC*.

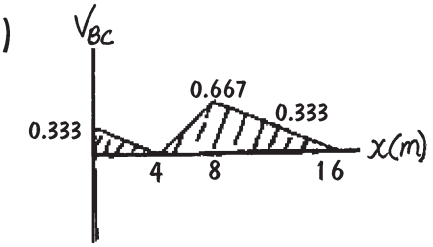



SOLUTION

(a)

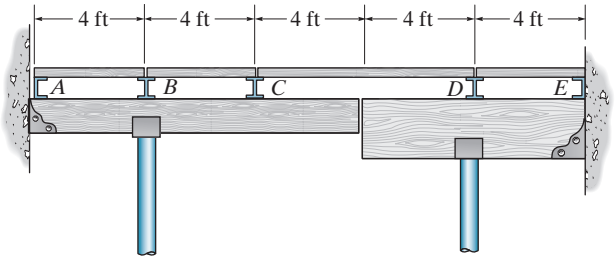



(b)



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**6–33.** A uniform live load of 300 lb/ft and a single concentrated live force of 2 k are to be placed on the floor beams. Determine (a) the maximum negative shear in panel $AB$, and (b) the maximum negative moment at $B$. Assume the supports at $A$ and $E$ are pins and the pipe columns only exert vertical reactions on the beams.




## SOLUTION

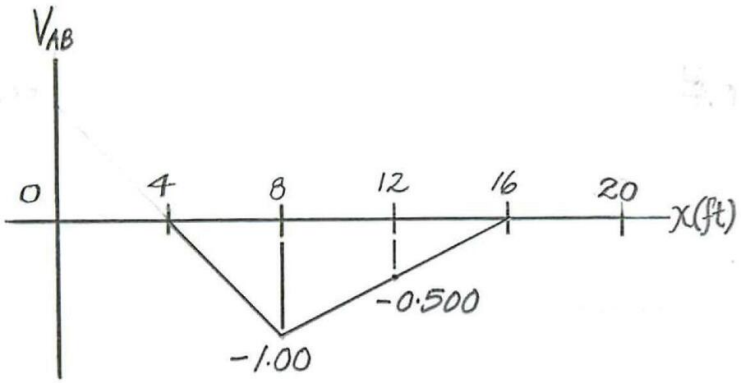
Referring to the influence line of the shear within panel $AB$ shown in Fig. $a$, the maximum negative shear within this panel is

$$(V_{AB})_{\max(-)} = 2(-1.00) + \left[\frac{1}{2}(16 - 4)(-1.00)\right](0.3) = -3.80\ \text{k} \quad \textbf{Ans.}$$

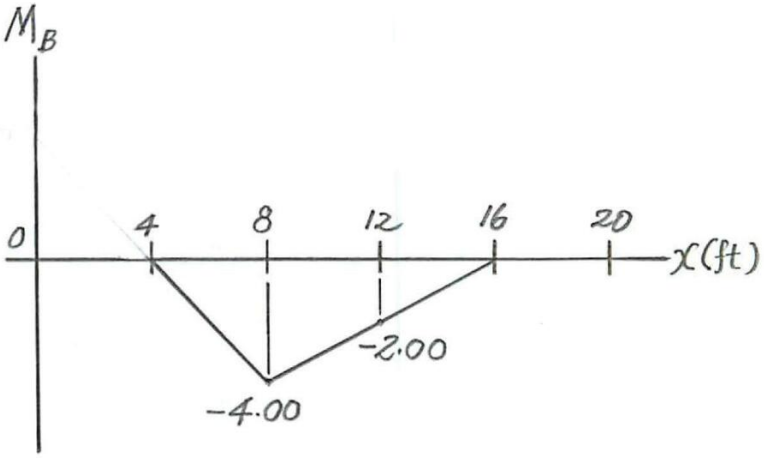
Referring to the influence line of moment at $B$, shown in Fig. $b$, the maximum negative moment at this point is

$$(M_B)_{\max(-)} = 2(-4.00) + \left[\frac{1}{2}(16 - 4)(-4.00)\right](0.3)$$

$$= -15.2\ \text{k} \cdot \text{ft} \quad \textbf{Ans.}$$




(a)

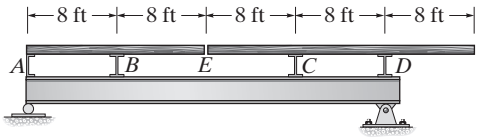



(b)

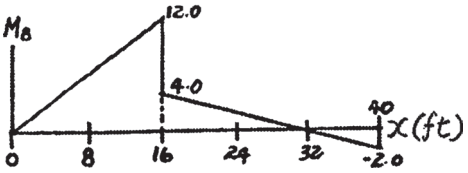
**Ans.**
$(V_{AB})_{\max(-)} = -3.80$ k;
$(M_B)_{\max(-)} = -15.2\ \text{k} \cdot \text{ft}$

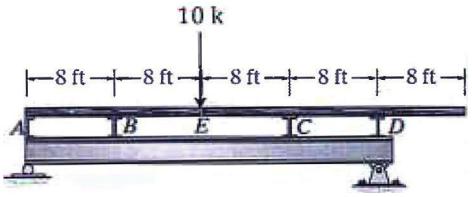
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**6–34.** Draw the influence line for the moment at $B$ in the girder. Determine the maximum positive live moment in the girder at $B$ if a single concentrated live force of 10 k moves across the top beams. Assume the supports for these beams can exert both upward and downward forces on the beams.




## SOLUTION



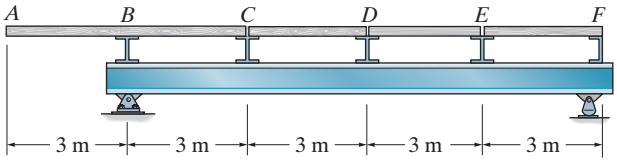




$(M_B)_{\max(+)} = 10(12) = 120\ \text{k} \cdot \text{ft}$ **Ans.**

**Ans.**
$(M_B)_{\max(+)} = 120\ \text{k} \cdot \text{ft}$

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**6–35.** Determine the maximum positive live shear in panel *CD* due to a uniform live load of 20 kN/m acting on the top beams.

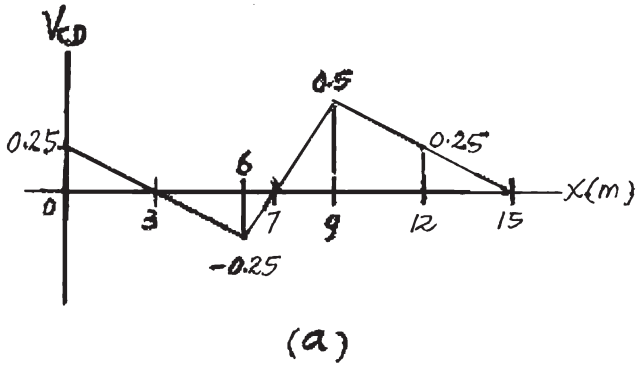



## SOLUTION

Referring to the influence line for the shear within panel *CD* shown in Fig. *a*, the maximum positive shear within this panel is

$$(V_{CD})_{\max(+)} = \left[\frac{1}{2}(3)(0.25)\right](20) + \left[\frac{1}{2}(15 - 7)(0.5)\right](20)$$

$$= 47.5 \text{ kN} \qquad \textbf{Ans.}$$

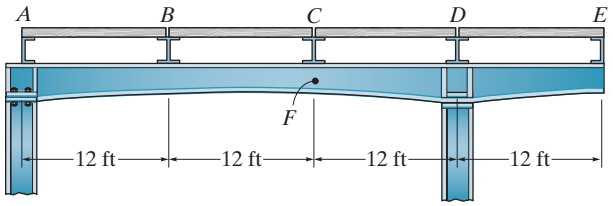



(a)

**Ans.**
$(V_{CD})_{\max(+)} = 47.5 \text{ kN}$

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***6–36.** A uniform live load of 3 k/ft and a single concentrated live force of 45 k are placed on the floor beams. If the beams also support a uniform dead load of 0.5 k/ft, determine (a) the maximum negative shear in panel *CD* of the girder and (b) the maximum positive moment in the girder at *F*.




## SOLUTION

Referring to the influence line for the shear within panel *CD* shown in Fig. *a*, the maximum shear within this panel is

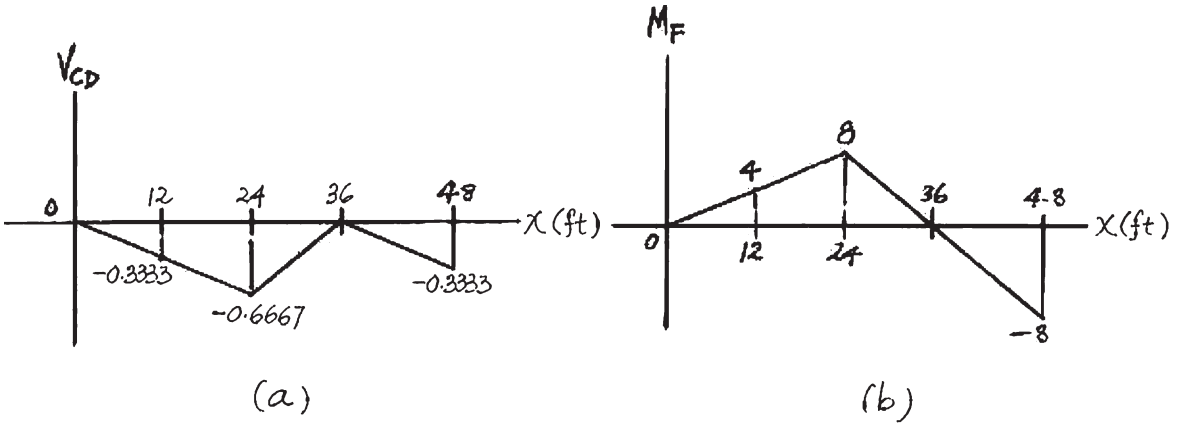
$$(V_{CD})_{\max(-)} = \left[\frac{1}{2}(36)(-0.6667)\right](3 + 0.5) + \left[\frac{1}{2}(48 - 36)(-0.3333)\right](3 + 0.5) + 45(-0.6667)$$

$$= 79.0 \text{ k} \qquad \textbf{Ans.}$$

Referring to the influence line for the moment at *F* shown in Fig. *b*, the maximum moment at this point is

$$(M_F)_{\max(+)} = \left[\frac{1}{2}(36)(8)\right](3 + 0.5) + \left[\frac{1}{2}(48 - 36)(-8)\right](0.5) + 45(8)$$

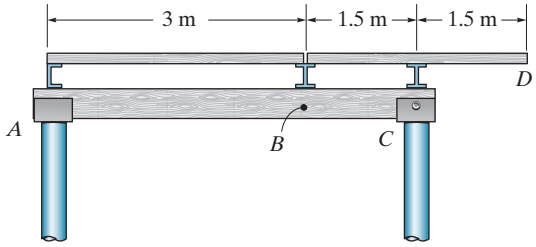
$$= 840 \text{ k} \cdot \text{ft} \qquad \textbf{Ans.}$$




**Ans.**
$(V_{CD})_{\max(-)} = 79.0$ k;
$(M_F)_{\max(+)} = 840$ k · ft

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**6–37.** A uniform live load of 1.75 kN/m and a single concentrated live force of 8 kN are placed on the floor beams. If the beams also support a uniform dead load of 250 N/m, determine (a) the maximum negative shear in panel *BC* of the girder and (b) the maximum positive moment at *B*.



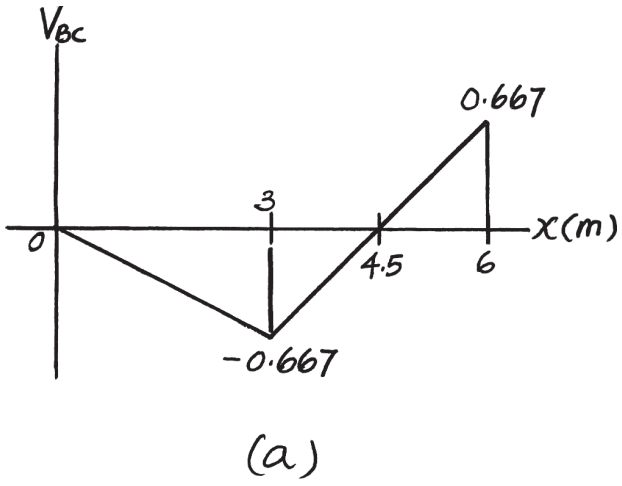

## SOLUTION

By referring to the influence line for the shear in panel *BC* shown in Fig. *a*, the maximum negative shear is

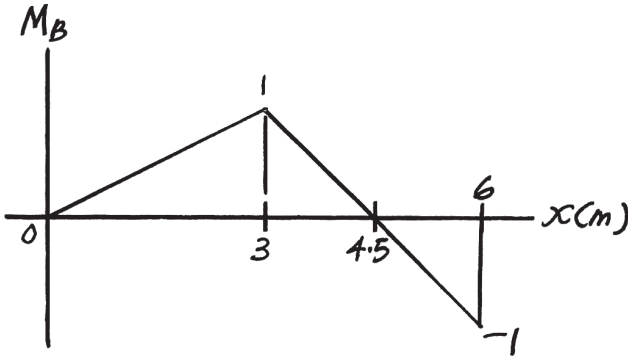
$$(V_{BC})_{\max(-)} = -0.6667(8) + \left[\frac{1}{2}(4.5 - 0)(-0.6667)\right](1.75 + 0.25) + \left[\frac{1}{2}(6 - 4.5)(0.6667)\right](0.25)$$
$$= -8.21 \text{ kN} \quad \textbf{Ans.}$$

By referring to the influence line for the moment at *B* shown in Fig. *b*, the maximum positive moment is

$$(M_B)_{\max(+)} = 1(8) + \left[\frac{1}{2}(4.5 - 0)(1)\right](1.75 + 0.25) + \left[\frac{1}{2}(6 - 4.5)(-1)\right](0.25)$$
$$= 12.3 \text{ kN} \cdot \text{m} \quad \textbf{Ans.}$$




(a)

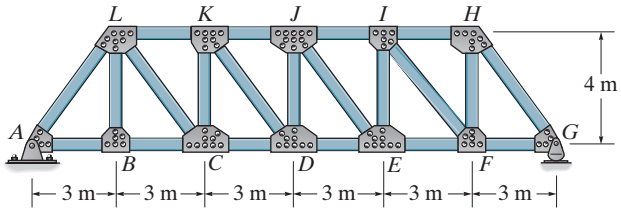



(b)

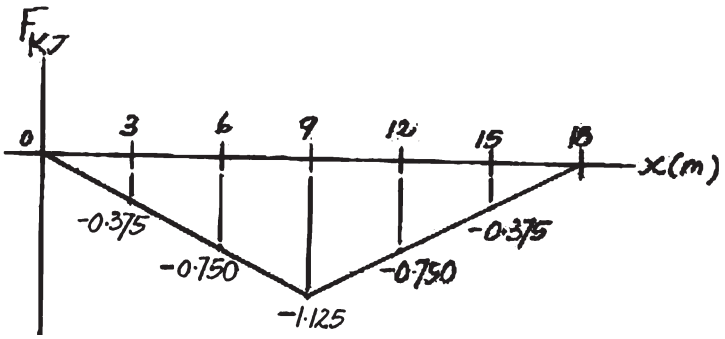
**Ans.**
$(V_{BC})_{\max(-)} = -8.21$ kN;
$(M_B)_{\max(+)} = 12.3$ kN · m

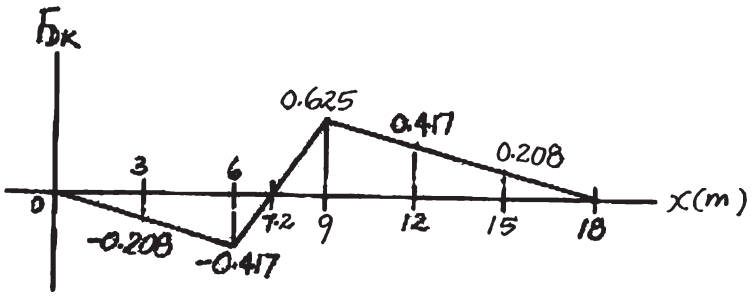
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**6–38.** Draw the influence line for the force in (a) member *KJ* and (b) member *DK*.




## SOLUTION

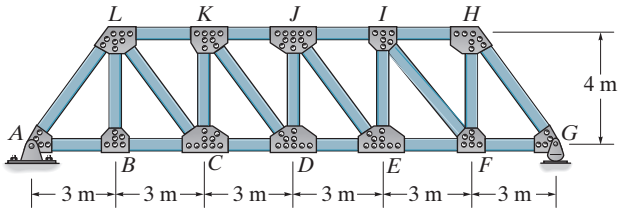


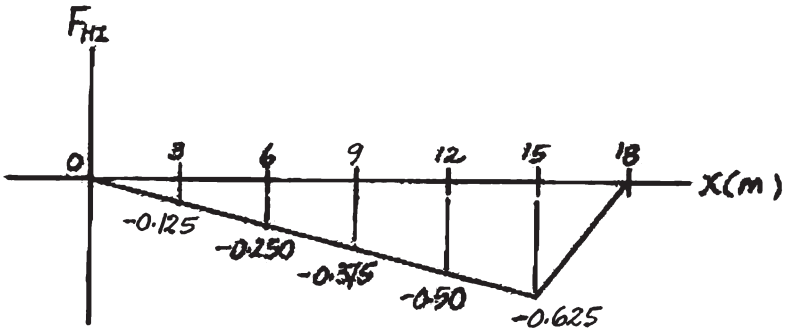

**Ans.**
$(F_{KJ})_{\max} = -1.125$
$(F_{DK})_{\max} = 0.625$

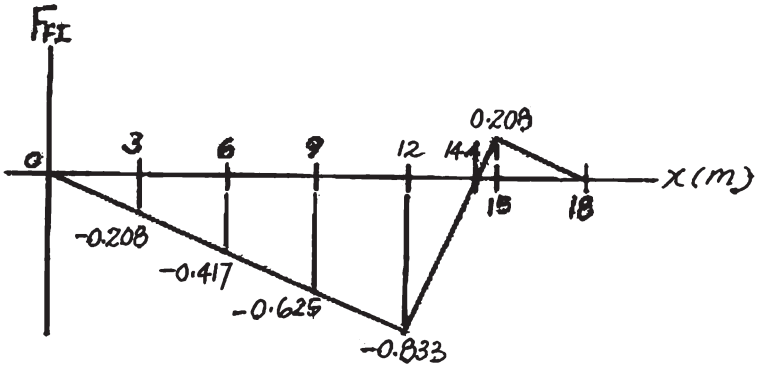
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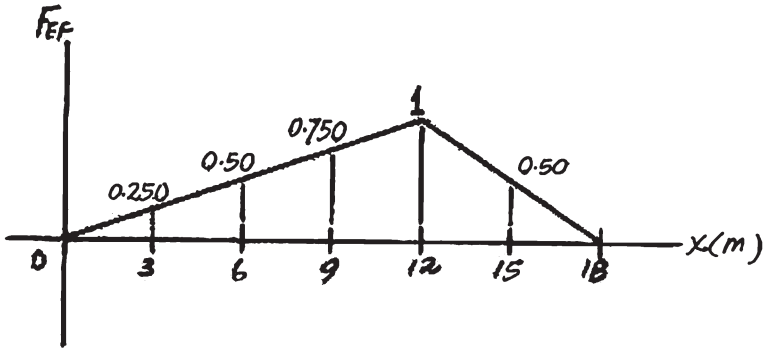
**6–39.** Draw the influence line for the force in (a) member *HI*, (b) member *FI*, and (c) member *EF*.




## SOLUTION



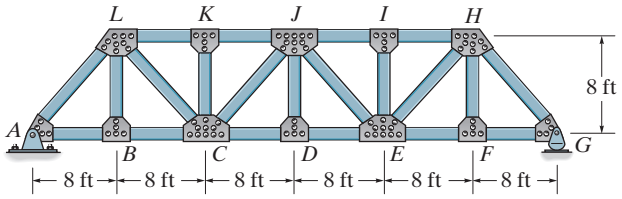



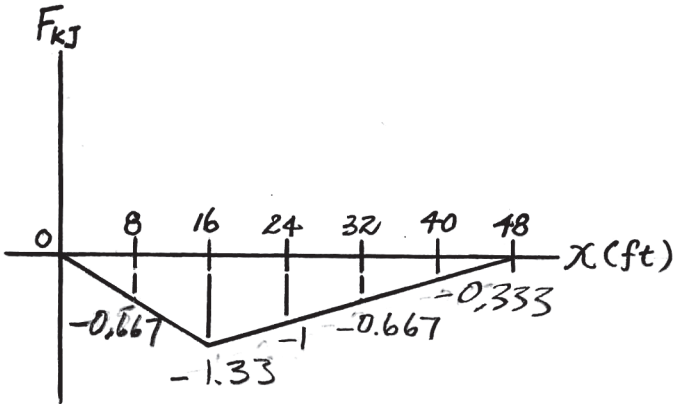

**Ans.**
$(F_{HI})_{max} = -0.625$
$(F_{FI})_{max} = -0.833$
$(F_{EF})_{max} = 1$

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***6–40.** Draw the influence line for the force in member *KJ*.

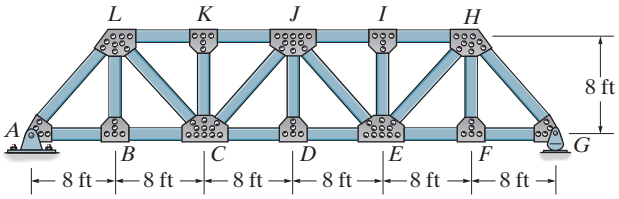



SOLUTION

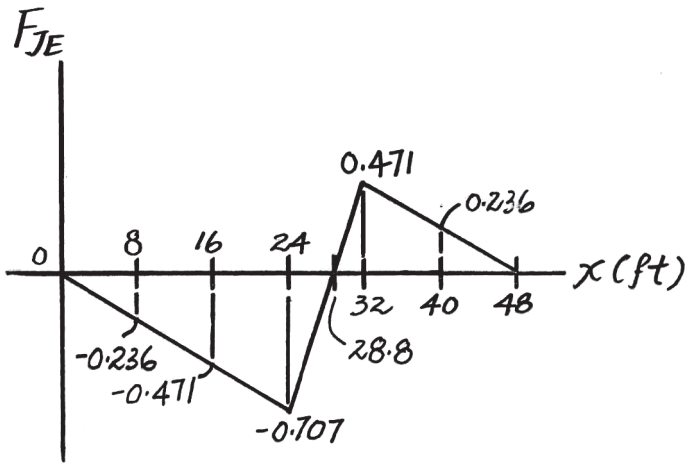


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**6–41.** Draw the influence line for the force in member *JE*.



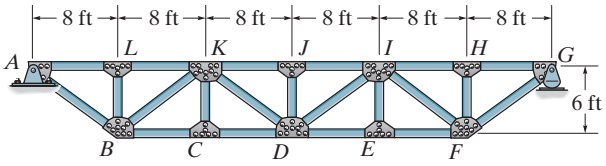

SOLUTION



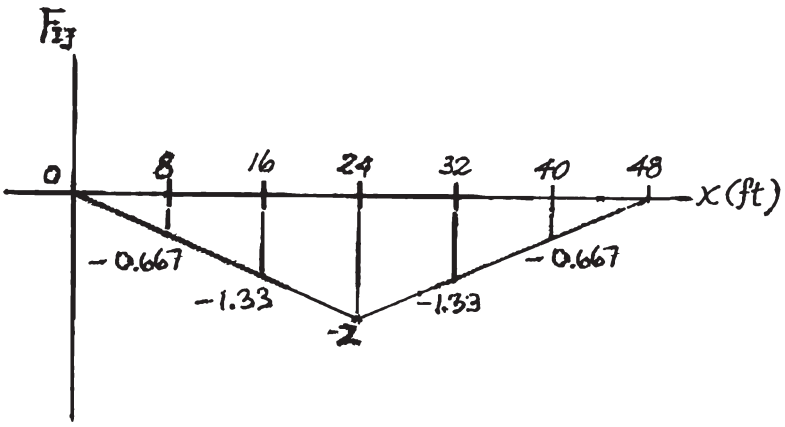

**Ans.**
$(F_{JE})_{\max} = -0.707$

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**6–42.** Draw the influence line for the force in member *IJ*.



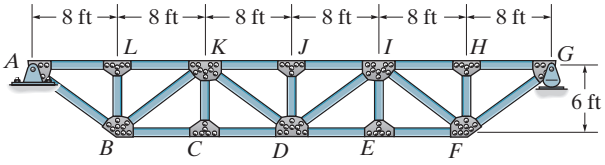

SOLUTION



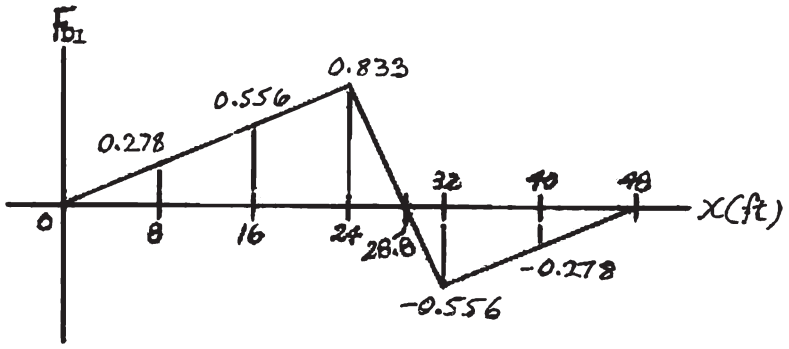

**Ans.**
$(F_{IJ})_{\max} = -2$

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**6–43.** Draw the influence line for the force in member *DI*.



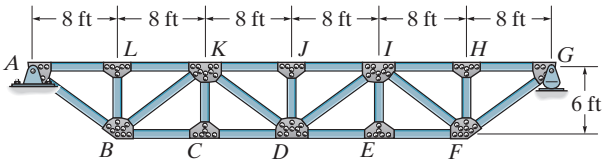

## SOLUTION



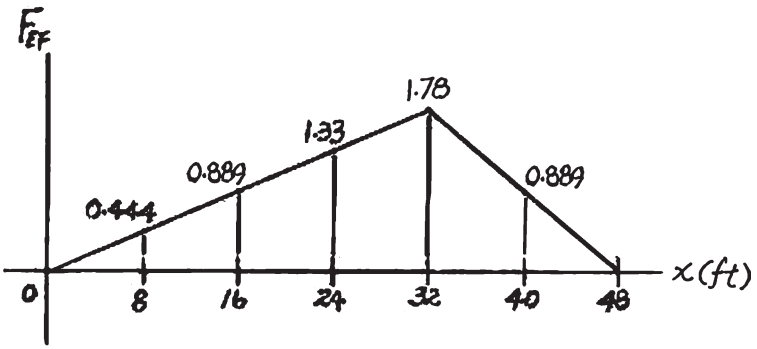

**Ans.**
$(F_{DI})_{\max} = 0.833$

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***6–44.** Draw the influence line for the force in member *EF*.

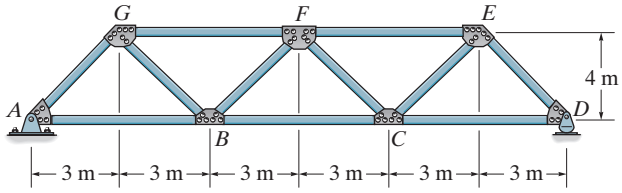



## SOLUTION

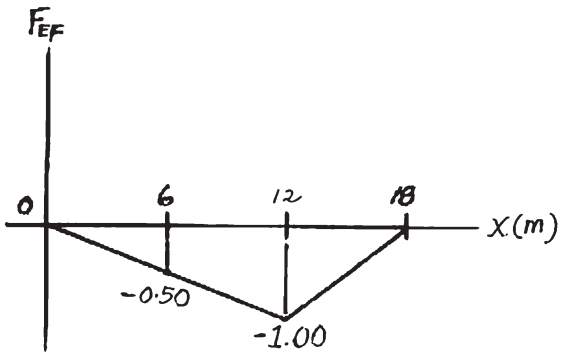


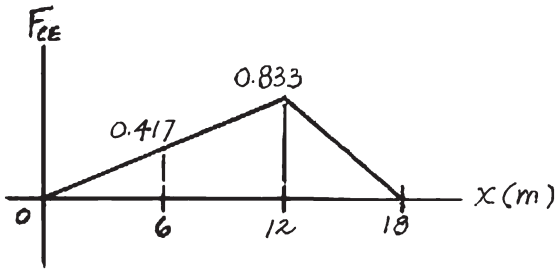
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**6–45.** Draw the influence line for the force in (a) member *EF* and (b) member *CE*.




## SOLUTION

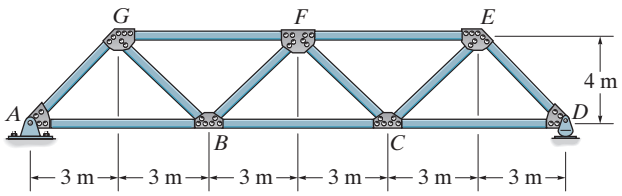


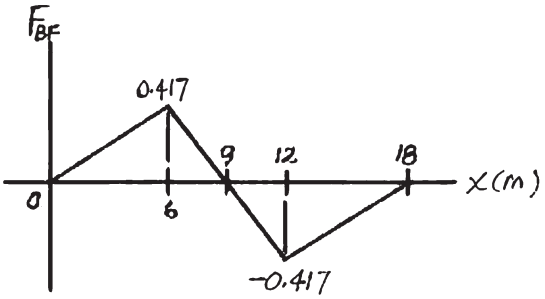

**Ans.**
$(F_{EF})_{max} = -1$
$(F_{CE})_{max} = 0.833$

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**6–46.** Draw the influence line for the force in member *BF*.



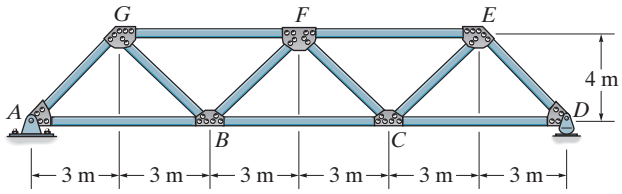

## SOLUTION



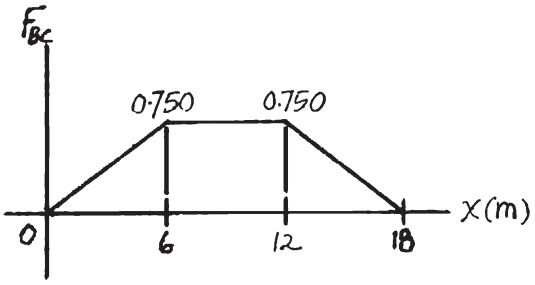

**Ans.**
$(F_{BF})_{max} = \pm 0.417$

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**6–47.** Draw the influence line for the force in member *BC*.



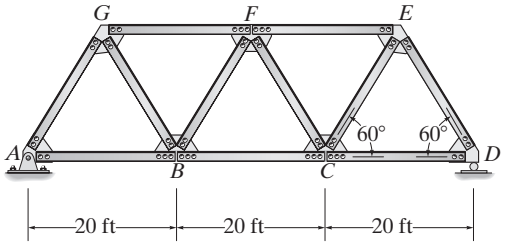

## SOLUTION



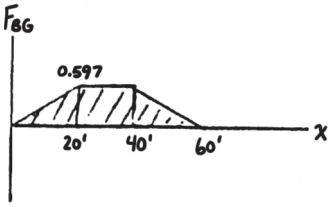

**Ans.**
$(F_{BC})_{\max} = 0.75$

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***6–48.** Draw the influence line for the force in member $BC$ of the Warren truss. Indicate numerical values for the peaks. All members have the same length.

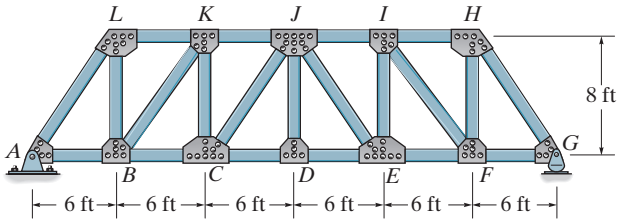



## SOLUTION

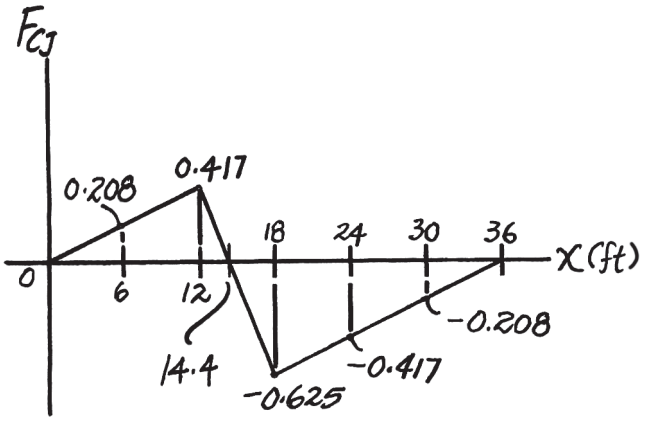


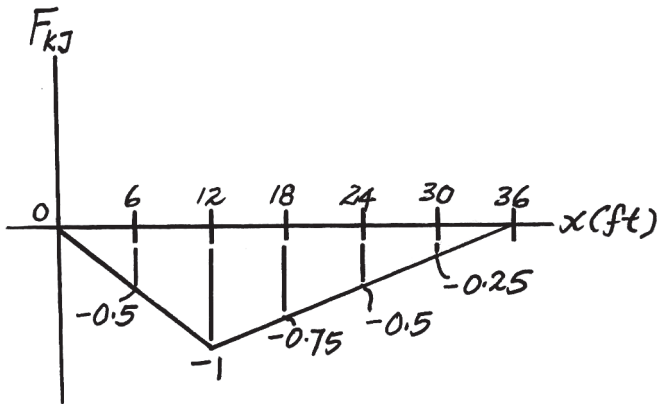
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**6–49.** Draw the influence line for the force in (a) member *KJ* and (b) member *CJ*.




SOLUTION

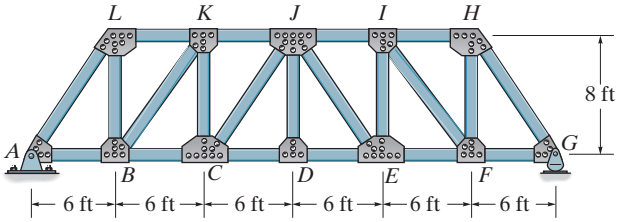


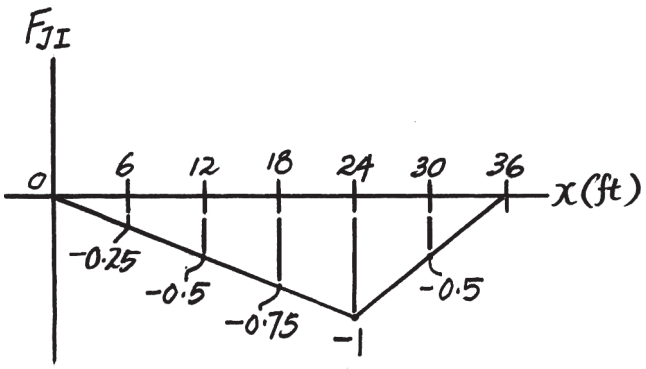

**Ans.**
$(F_{KJ})_{\max} = -1$
$(F_{CJ})_{\max} = -0.625$

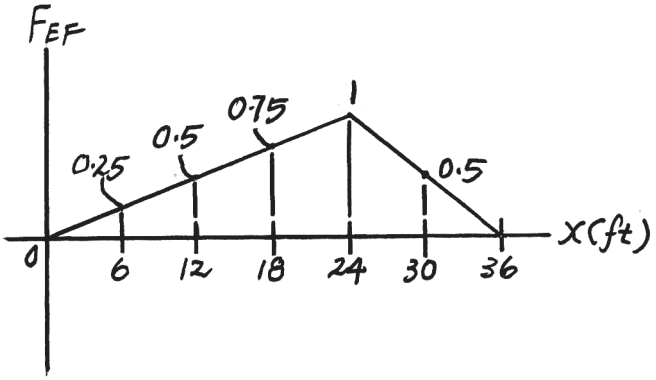
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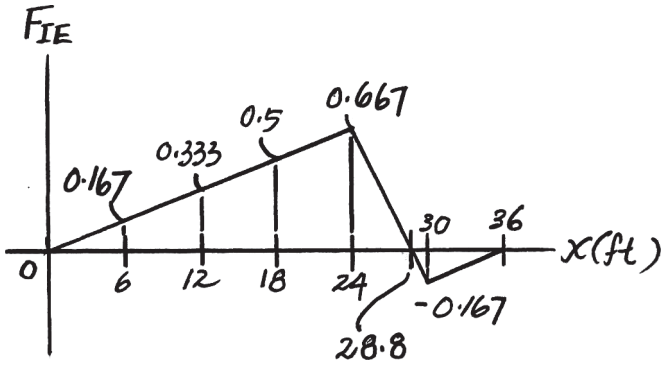
**6–50.** Draw the influence line for the force in (a) member *JI*, (b) member *IE*, and (c) member *EF*.




SOLUTION



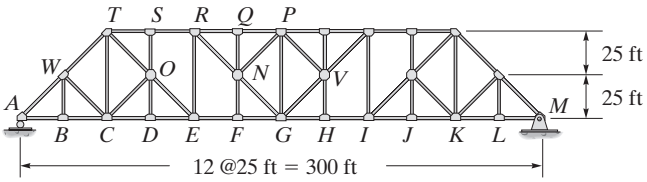



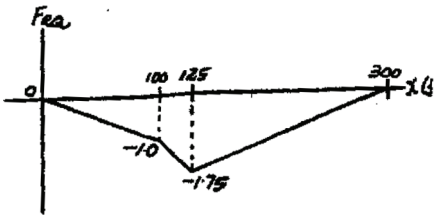

**Ans.**
$(F_{JI})_{\max} = -1$
$(F_{IE})_{\max} = 0.667$
$(F_{EF})_{\max} = 1$

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**6–51.** Draw the influence line for the force in member *RQ* of the Baltimore truss.



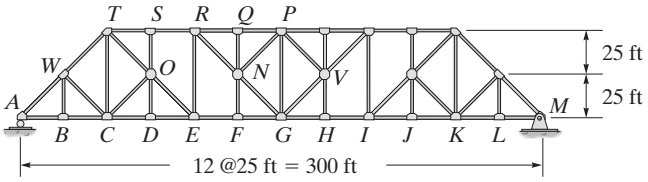

## SOLUTION



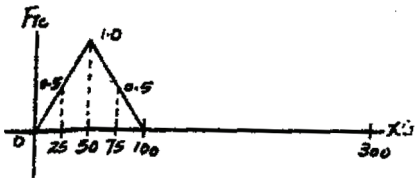

**Ans.**
$(F_{RQ})_{\max} = -1.75$

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***6–52.** Draw the influence line for the force in member *TC* of the Baltimore truss.

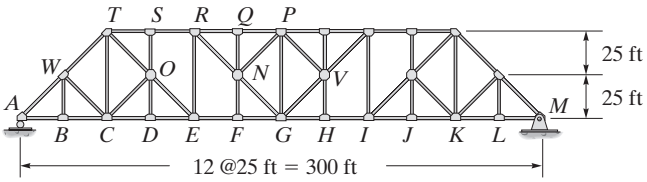



## SOLUTION

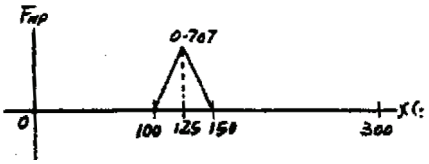


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**6–53.** Draw the influence line for the force in member *NP* of the Baltimore truss.



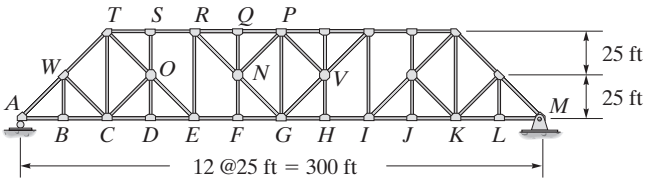

## SOLUTION



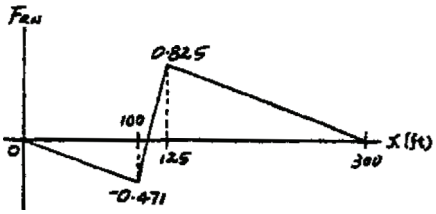

**Ans.**
$(F_{NP})_{\max} = 0.707$

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**6–54.** Draw the influence line for the force in member *RN* of the Baltimore truss.



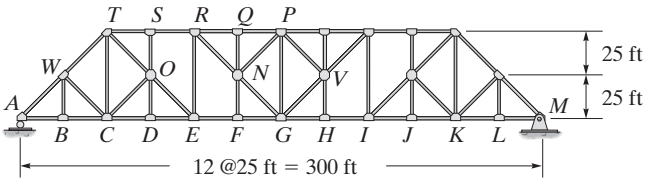

## SOLUTION



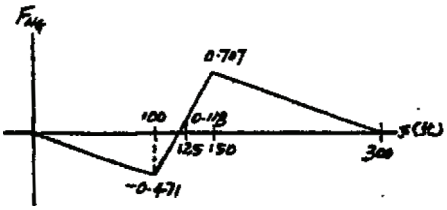

**Ans.**
$(F_{RN})_{max} = 0.825$

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**6–55.** Draw the influence line for the force in member *NG* of the Baltimore truss.



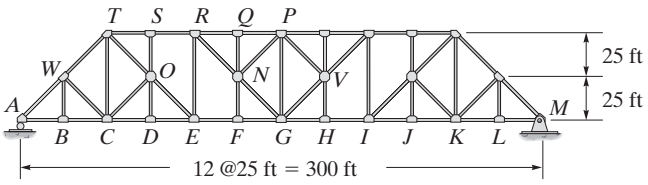

## SOLUTION



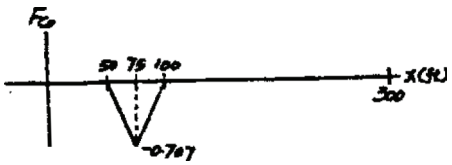

**Ans.**
$(F_{NG})_{\max} = 0.707$

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***6–56.** Draw the influence line for the force in member *CO* of the Baltimore truss.

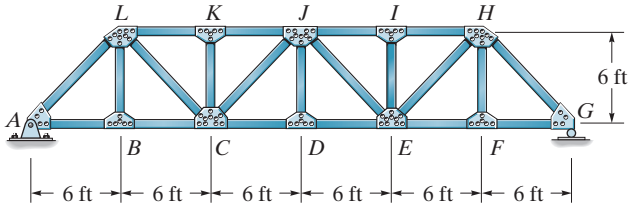



## SOLUTION

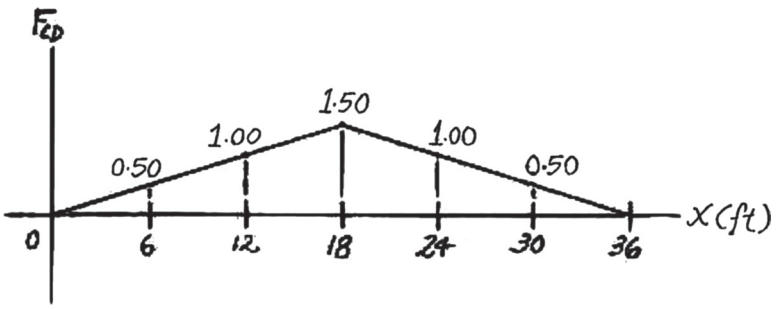


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**6–57.** Draw the influence line for the force in member *CD*.



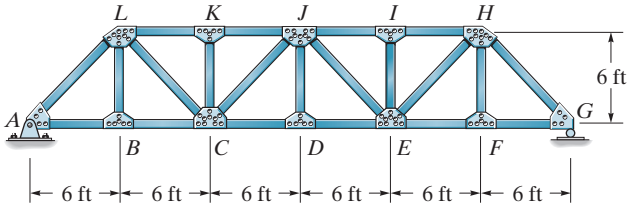

SOLUTION



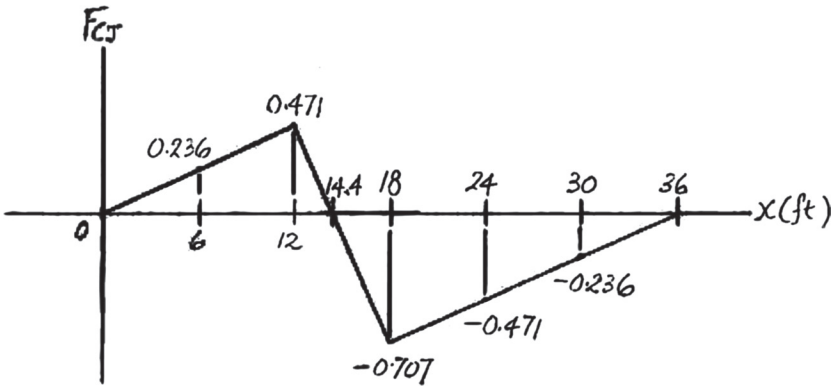

**Ans.**
$(F_{CD})_{max} = 1.50$

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**6–58.** Draw the influence line for the force in member *CJ*.



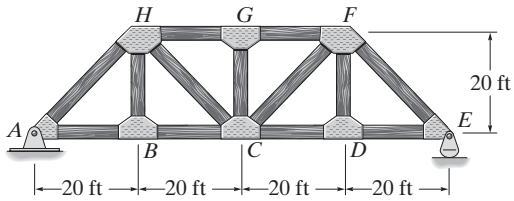

## SOLUTION




**Ans.**
$(F_{CJ})_{\max} = -0.707$

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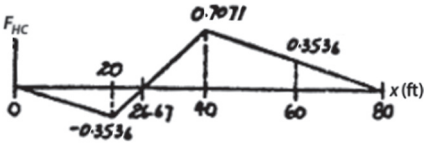
**6–59.** Draw the influence line for the force in member *HC*, then determine the maximum live force (tension or compression) that can be developed in this member due to a uniform live load of 800 lb/ft that acts on the bridge deck along the bottom cord of the truss.

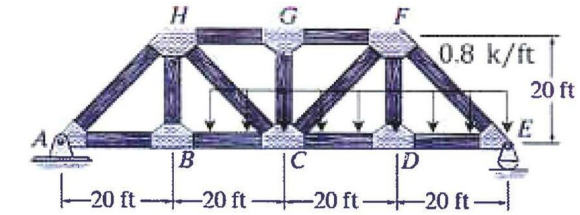


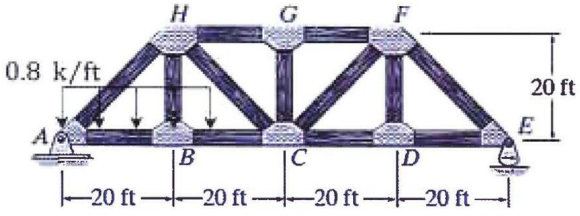

## SOLUTION

$$(F_{HC})_{\max\,(T)} = 0.8\left(\frac{1}{2}\right)(0.7071)(53.33) = 15.1\text{ k (T)}$$ **Ans.**

$$(F_{HC})_{\max\,(C)} = 0.8\left(\frac{1}{2}\right)(-0.3536)(26.67) = -3.77\text{ k} = 3.77\text{ k (C)}$$ **Ans.**



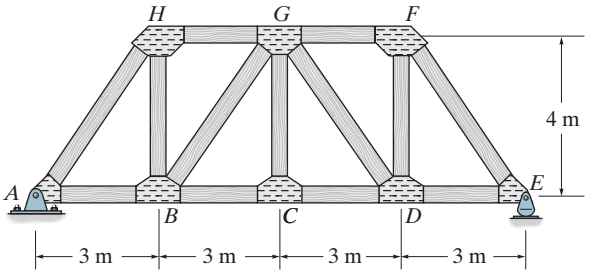

**Ans.**

$(F_{HC})_{\max\,(T)} = 15.1\text{ k (T)};$

$(F_{HC})_{\max\,(C)} = 3.77\text{ k (C)}$

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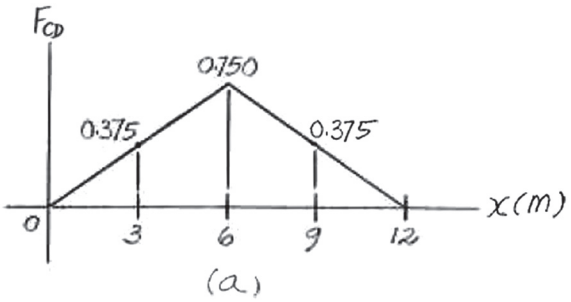
***6–60.** Draw the influence line for the force in member *CD*, and then determine the maximum force (tension or compression) that can be developed in this member due to a uniform live load of 24 kN/m which acts along the bottom cord of the truss.




## SOLUTION

Referring to the influence line for the force in member *CD* shown in Fig. *a*, we notice that member *CD* is subjected to only tensile force. Thus, the maximum force in this member is

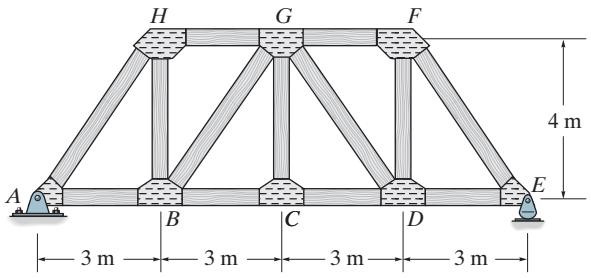
$$(F_{CD})_{\max} = \left[\frac{1}{2}(12)(0.75)\right](24) = 108 \text{ kN (T)} \qquad \textbf{Ans.}$$




**Ans.**
$(F_{CD})_{\max} = 108 \text{ kN (T)}$

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**6–61.** Draw the influence line for the force in member $DG$, and then determine the maximum force (tension or compression) that can be developed in this member due to a uniform live load of 24 kN/m which is transmitted to the truss along the bottom cord.



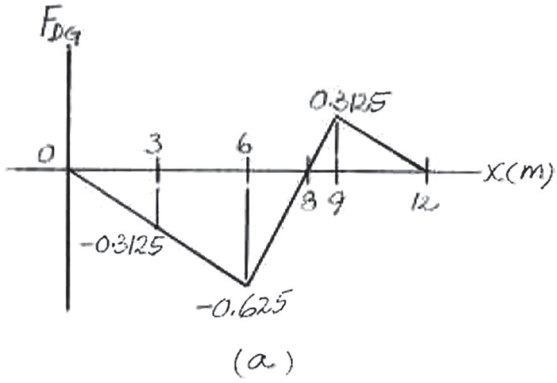

## SOLUTION

Referring to the influence line for the force in member $DG$ shown in Fig. $a$, we notice that member $DG$ could be subjected to tensile or compression force. The maximum tensile force and compressive force are

$$(F_{DG})_{\max\,(T)} = \left[\frac{1}{2}(12.8)(0.3125)\right](24) = 15.0\ \text{kN (T)}$$

$$(F_{DG})_{\max\,(C)} = \left[\frac{1}{2}(8)(-0.625)\right](24) = -60.0\ \text{kN} = 60.0\ \text{kN (C) (Max!)}$$

**Ans.**

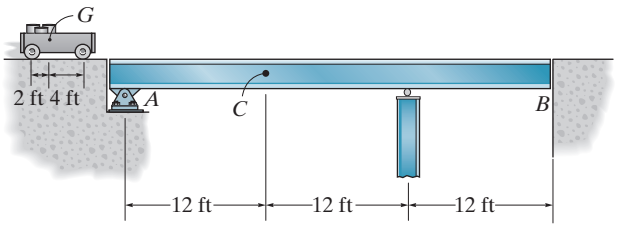



**Ans.**

$$(F_{DG})_{\max} = 60.0\ \text{kN (C)}$$

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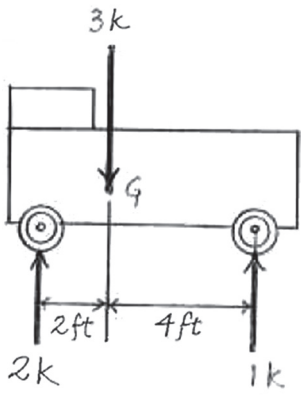
**6–62.** Determine the maximum moment at point $C$ on the single girder caused by the moving dolly that has a weight of 3 k and a center of gravity at $G$.



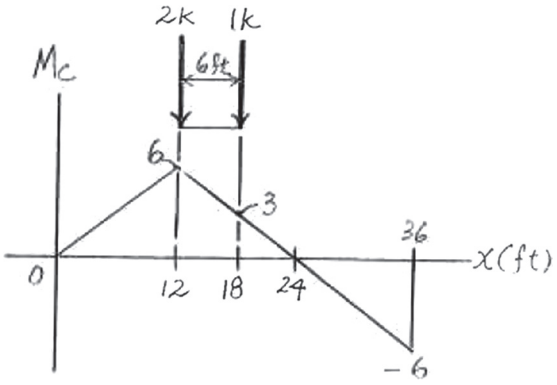

## SOLUTION

The vertical reactions of the wheels on the girder are shown in Fig. *a*. The maximum moment at point $C$ occurs when the moving loads are at the position shown in Fig. *b*.

$(M_C)_{\max(+)} = 6(2) + 3(1) = 15.0\ \text{k}\cdot\text{ft}$ **Ans.**




(a)

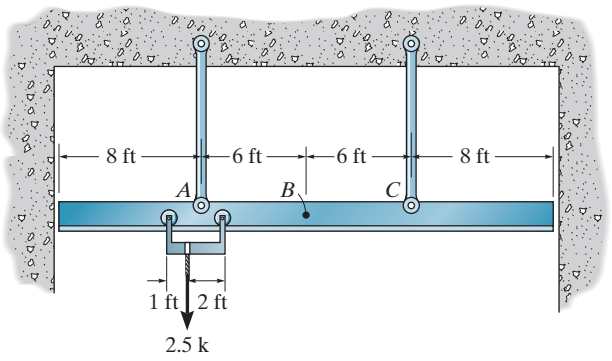



(b)

**Ans.**
$(M_C)_{\max(+)} = 15.0\ \text{k}\cdot\text{ft}$

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**6–63.** Determine the maximum moment in the suspended rail at point *B* if the rail supports the load of 2.5 k on the trolley.




## SOLUTION

Check maximum positive moment:

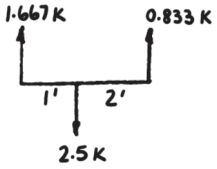
$$\frac{h}{3} = \frac{3}{6}; \qquad h = 1.5 \text{ ft}$$

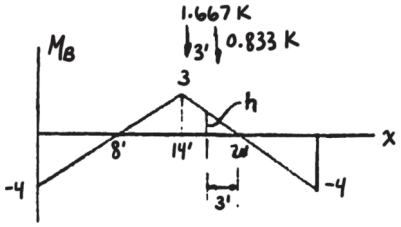
$$(M_B)_{\max} = 1.667(3) + (0.833)(1.5) = 6.25 \text{ k} \cdot \text{ft}$$

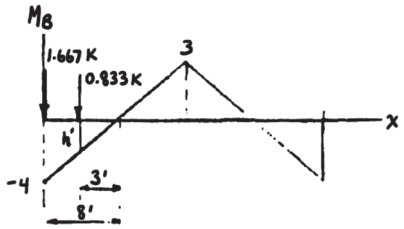
Check maximum negative moment:

$$\frac{h}{5} = \frac{4}{8}; \qquad h = 2.5 \text{ ft}$$

$$(M_B)_{\max} = 1.667(-4) + (0.833)(-2.5) = -8.75 \text{ k} \cdot \text{ft}$$ **Ans.**



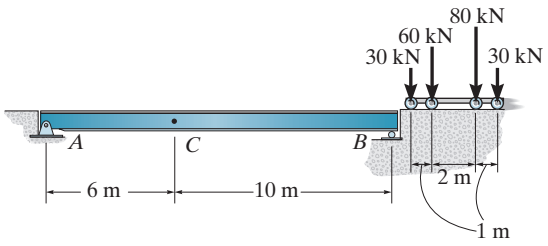




**Ans.**
$(M_B)_{\max} = -8.75 \text{ k} \cdot \text{ft}$

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***6–64.** Determine the maximum live moment at $C$ caused by the moving loads.




## SOLUTION

Moving the 60-kN force 2 m to the left of point $C$, the change in moment is

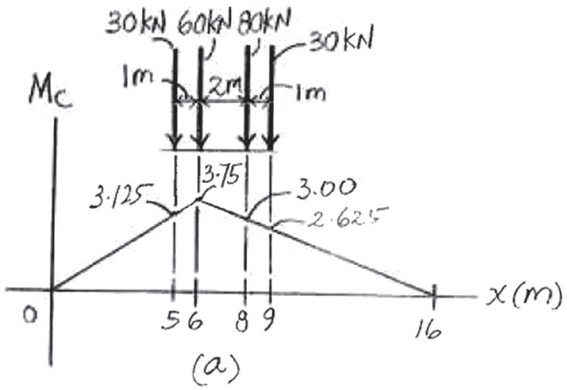
$$\Delta M = 30\left(-\frac{3.75}{6}\right)(2) + 60\left(-\frac{3.75}{6}\right)(2) + 80\left(\frac{3.75}{10}\right)(2) + 30\left(\frac{3.75}{10}\right)(2)$$

$$= -30\ \text{kN}\cdot\text{m}$$

Since $\Delta M$ is negative, the case where the 60-kN force at point $C$ will generate the maximum positive moment, Fig. *a*.

$$(M_C)_{\max(+)} = 3.125(30) + 3.75(60) + 3.00(80) + 30(2.625)$$

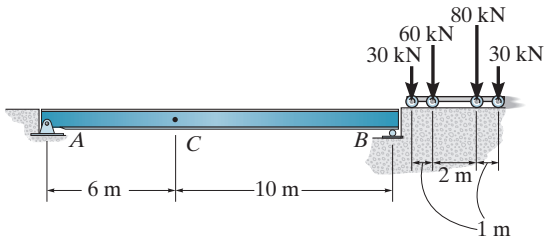
$$= 637.5\ \text{kN}\cdot\text{m}$$ **Ans.**




**Ans.**
$(M_C)_{\max(+)} = 637.5\ \text{kN}\cdot\text{m}$

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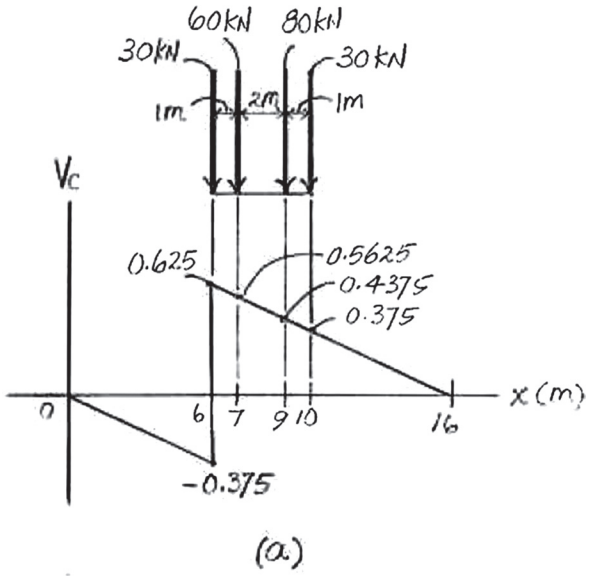
**6–65.** Determine the maximum live shear at *C* caused by the moving loads.




## SOLUTION

By inspecting the influence line for the shear at *C*, the maximum shear at this point is positive and occurs when the moving loads are at the position shown in Fig. *a*.

$$(V_C)_{\max(+)} = 30(0.625) + 60(0.5625) + 80(0.4375) + 30(0.375)$$
$$= 98.75 \text{ kN} \quad \textbf{Ans.}$$

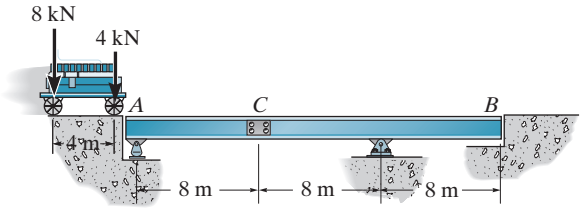



(a)

**Ans.**
$(V_C)_{\max(+)} = 98.75$ kN

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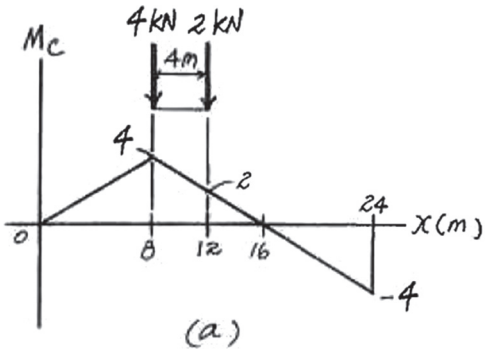
**6–66.** Determine the maximum positive moment at the splice $C$ on the side girder caused by the moving load which travels along the center of the bridge.




## SOLUTION

The maximum positive moment at point $C$ occurs when the moving loads are at the position shown in Fig. *a*.

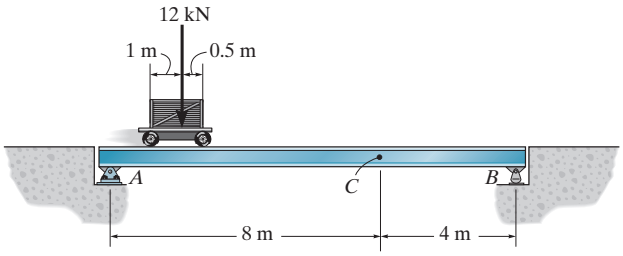
$(M_C)_{\max(+)} = 4(4) + 2(2) = 20.0\ \text{kN}\cdot\text{m}$ **Ans.**




**Ans.**
$(Mc)_{max} = 20.0\ \text{kN}\cdot\text{m}$

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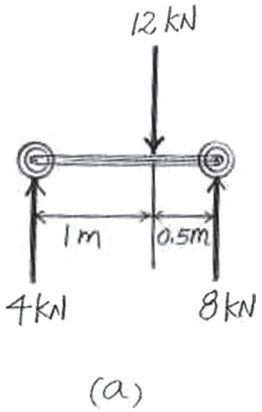
**6–67.** Determine the maximum moment at $C$ caused by the moving load.



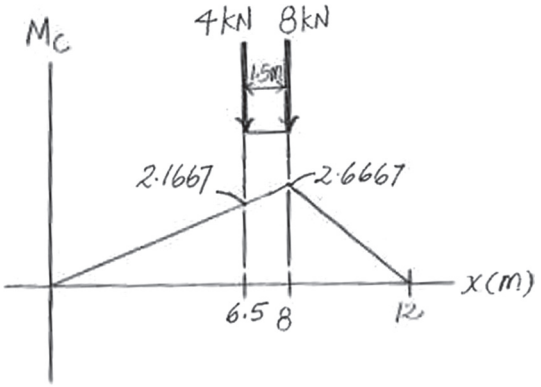

## SOLUTION

The vertical reactions of the wheels on the beam are shown in Fig. *a*. The maximum moment at point $C$ occurs when the moving loads are at the position shown in Fig. *b*.

$(M_C)_{\max(+)} = 8(2.6667) + 4(2.1667) = 30.0\ \text{kN}\cdot\text{m}$ **Ans.**




(a)

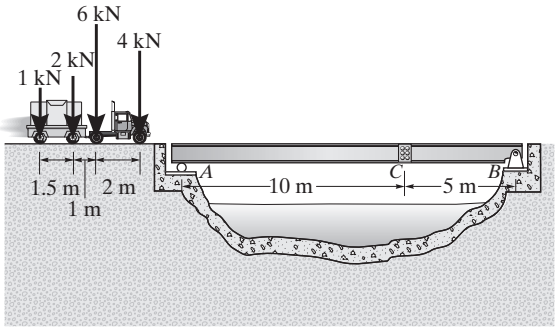



(b)

**Ans.**
$(M_C)_{\max(+)} = 30.0\ \text{kN}\cdot\text{m}$

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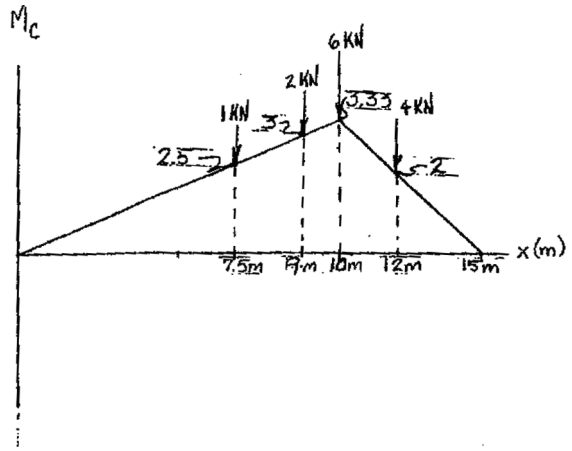
***6–68.** The truck and trailer exerts the wheel reactions shown on the deck of the girder bridge. Determine the largest moment it exerts in the splice at *C*. Assume the truck travels left to right along the *center* of the deck, and therefore transfers *half* of the load shown to each of the two side girders. Assume the splice is a fixed connection and, like the girder, can support both shear and moment.




## SOLUTION

Worst case is 6 kN at $P = 10$ m.

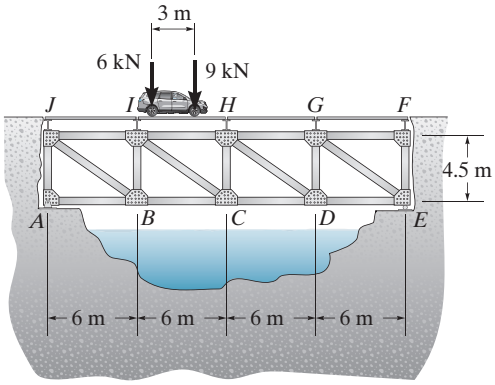
$$(M_C)_{\max} = \frac{1}{2}\Big[1\text{ kN}(2.5) + 2\text{ kN}(3) + 6\text{ kN}(3.33) + 4\text{ kN}(2)\Big]$$
$$= 18.25\text{ kN}\cdot\text{m} \qquad \textbf{Ans.}$$




**Ans.**
$(M_C)_{\max} = 18.25\text{ kN}\cdot\text{m}$

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**6–69.** Draw the influence line for the force in member *CD* of the bridge truss. Determine the maximum live force (tension or compression) that can be developed in the member due to a 15-kN truck having the wheel loads shown. Assume the truck can travel in *either direction* along the *center* of the deck, so that *half* the load shown is transferred to each of the two side trusses. Also assume the members are pin connected at the gusset plates.

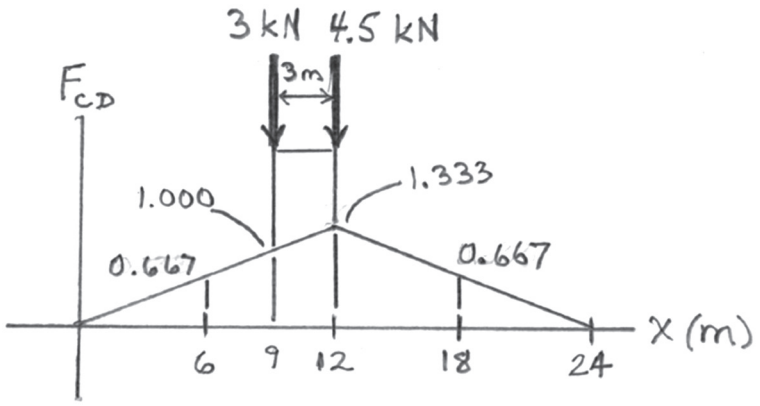



## SOLUTION

Refering to the influence for the force in member *CD* shown in Fig. *a*, we notice that member *CD* will be subjected to tensile force only, and the maximum occurs when the moving load, which is half that shown in the problem figure, is at the position shown.

$(F_{CD})_{\max} = 3(1.000) + 4.5(1.333) = 9.00\ \text{kN}(T)$ **Ans.**

By symmetry, the result would be the same if the truck were traveling in the other direction.

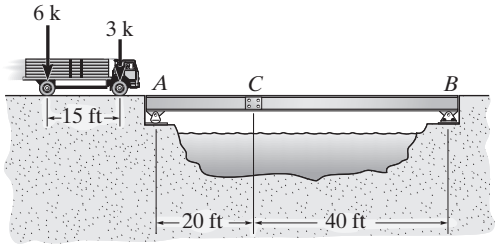



(a)

**Ans.**
$(F_{CD})_{\max} = 9.00\ \text{kN (T)}$

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**6–70.** The 9-k truck exerts the wheel reactions shown on the deck of a girder bridge. Determine (a) the largest live shear it creates in the splice at $C$, and (b) the largest moment it exerts in the splice. Assume the truck travels in *either direction* along the *center* of the deck, and therefore transfers *half* of the load shown to each of the two side girders. Assume the splice is a fixed connection and, like the girder, can support both shear and moment.

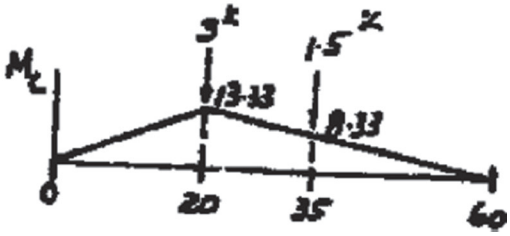



## SOLUTION

(a) $(V_C)_{\max} = 3(0.6667) + 1.5(0.4167) = 2.62\ \text{k}$ **Ans.**

(b) $(M_C)_{\max} = 3(13.33) + 1.5(8.333) = 52.5\ \text{k} \cdot \text{ft}$ **Ans.**

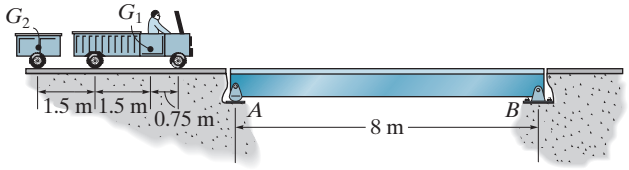



**Ans.**
$(V_C)_{\max} = 2.62\ \text{k}$;
$(M_C)_{\max} = 52.5\ \text{k} \cdot \text{ft}$

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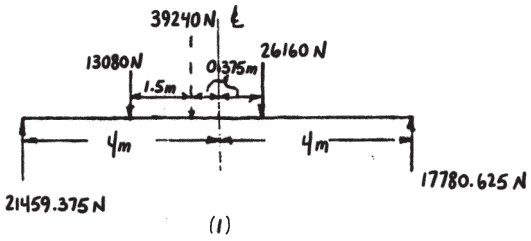
**6–71.** The truck has a mass of 4 Mg and mass center at $G_1$, and the trailer has a mass of 1 Mg and mass center at $G_2$. Determine the absolute maximum live moment developed in the bridge.

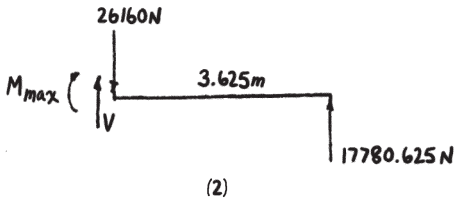



## SOLUTION

Placement is shown in FBD (1). Using segment (2):

$M_{max} = 17\,780.625(3.625) = 64.5\ \text{kN} \cdot \text{m}$ **Ans.**

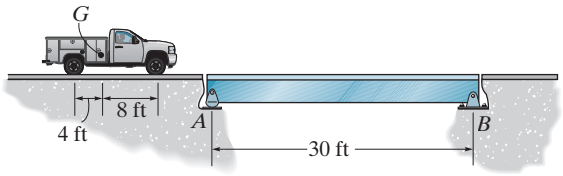



**Ans.**
$M_{\text{abs}}_{\text{max}} = 64.5\ \text{kN} \cdot \text{m}$

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***6–72.** The truck has a weight of 12 k and mass center at $G$. Determine the absolute maximum live moment developed in the bridge.



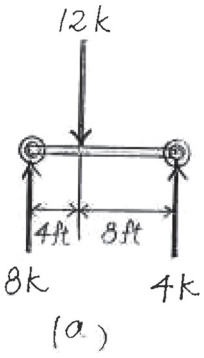

## SOLUTION

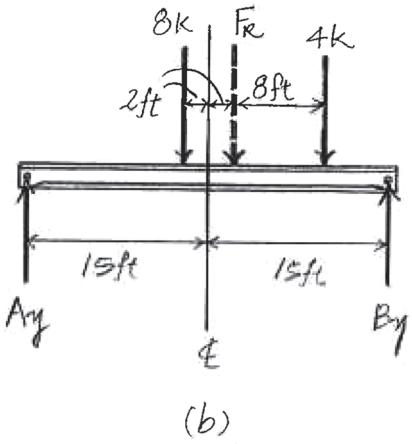
The vertical reactions of the wheels on the girder are shown in Fig. *a*. By inspection, the absolute maximum moment occurs under the 8-k force, Fig. *b*.

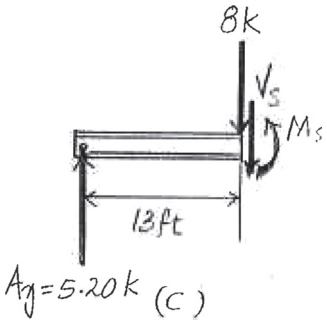
$\circlearrowleft + \Sigma M_B = 0; \quad -A_y(30) + 8(17) + 4(5) = 0 \quad A_y = 5.20\ \text{k}$

Referring to Fig. *c*,

$\circlearrowleft + \Sigma M_S = 0; \quad M_S - 5.20(13) = 0 \quad M_S = 67.6\ \text{k} \cdot \text{ft}$ (Abs. Max.) **Ans.**



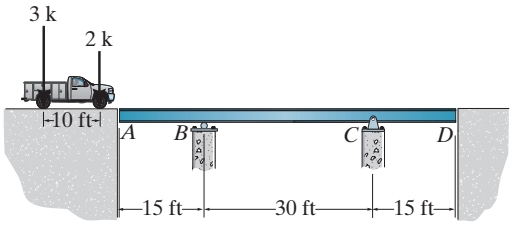




**Ans.**

$M_{\substack{\text{abs}\\\text{max}}} = 67.6\ \text{k} \cdot \text{ft}$

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**6–73.** Determine the absolute maximum live moment in the girder bridge due to the loading shown. The load is applied directly to the girder.




## SOLUTION

Referring to Fig. *a*, the location of $\boldsymbol{F}_R$ for the moving load is

$+\downarrow F_R = \Sigma F_y; \quad F_R = 3 + 2 = 5\ \text{k}$

$\circlearrowleft + F_R\bar{x} = \Sigma M_D; \quad -5\bar{x} = -2(10) \quad \bar{x} = 4\ \text{ft}$

By observation, the maximum positive moment occurs under the 3-k force, Fig. *b*.

$\circlearrowleft + \Sigma M_C = 0; \quad 2(7) + 3(17) - B_y(30) = 0 \quad B_y = 2.1667\ \text{k}$

Referring to Fig. *c*,

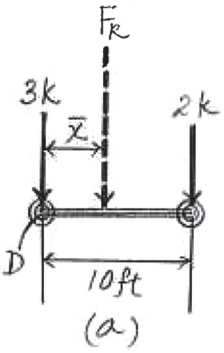
$\circlearrowleft + \Sigma M_S = 0; \quad M_{S(+)} - 2.1667(13) = 0$

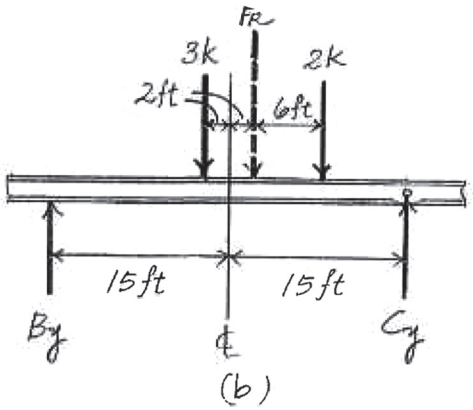
$M_{S(+)} = 28.17\ \text{k} \cdot \text{ft}$

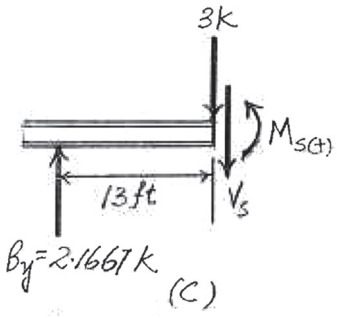
The maximum negative moment occurs when the 3-k force of the moving load is at the edge of the overhang, Fig. *d*.

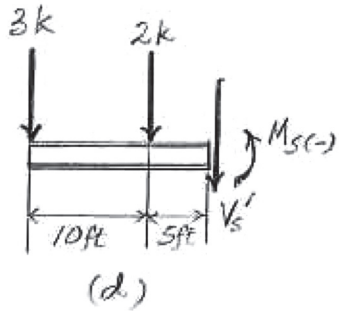
$\circlearrowleft + \Sigma M_S = 0; \quad M_{S(-)} + 2(5) + 3(15) = 0$

$M_{S(-)} = -55.0\ \text{k} \cdot \text{ft}$ (Abs. Max.) **Ans.**



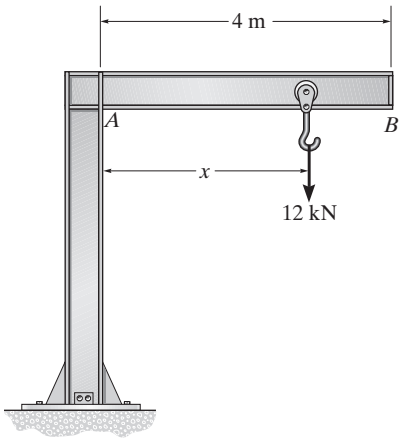




**Ans.**

$M_{\text{abs}}_{\text{max}} = -55.0\ \text{k} \cdot \text{ft}$

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**6–74.** Determine the absolute maximum live shear and absolute maximum live moment in the jib beam $AB$ due to the crane loading. The end constraints require $0.1\text{ m} \leq x \leq 3.9\text{ m}$.




## SOLUTION

$V_{\max} = +12\text{ kN}$ **Ans.**

(position at $0.1\text{ m} \leq x \leq 3.9\text{ m}$)

$M_{\max} = -12(3.9) = -46.8\text{ kN}\cdot\text{m}$ **Ans.**

(position at $x = 3.9\text{ m}$)

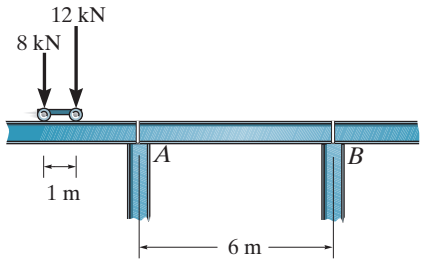
**Ans.**

$V_{\underset{\max}{\text{abs}}} = +12\text{ kN};$

$M_{\underset{\max}{\text{abs}}} = -46.8\text{ kN}\cdot\text{m}$

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**6–75.** The maximum wheel loadings for the wheels of a crane that is used in an industrial building are given. The crane travels along the runway girders that are simply supported on columns. Determine (a) the absolute maximum shear in intermediate girder $AB$, and (b) the absolute maximum moment in the girder.



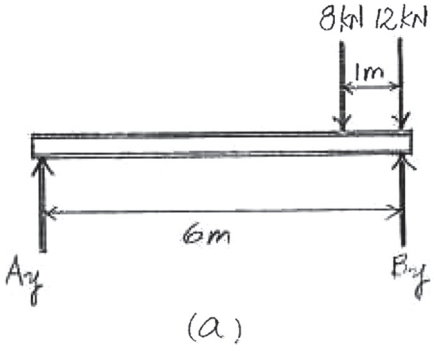

## SOLUTION

The absolute maximum shear occurs when the moving loads are positioned with the 12-kN force just to the left of the support at $B$, Fig. $a$.

$\circlearrowleft +\Sigma M_A = 0; \quad B_y(6) - 8(5) - 12(6) = 0 \quad B_y = 18.67\ \text{kN}$

Therefore, the absolute maximum shear stress is,

$$V_{\text{max}}^{\text{abs}} = B_y = 18.67\ \text{kN} = 18.7\ \text{kN} \qquad \textbf{Ans.}$$




(a)

Referring to Fig. $b$, the location of $\boldsymbol{F}_R$ for the moving load is

$+\downarrow F_R = \Sigma F_y; \quad F_R = 8 + 12 = 20\ \text{kN}$

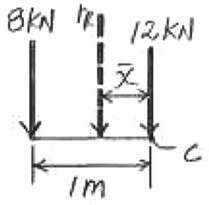
$\circlearrowleft +F_R\bar{x} = \Sigma M_C; \quad 20\bar{x} = 8(1) \quad \bar{x} = 0.4\ \text{m}$

By observation, the absolute maximum moment occurs under the 12-kN force, Fig. $c$.

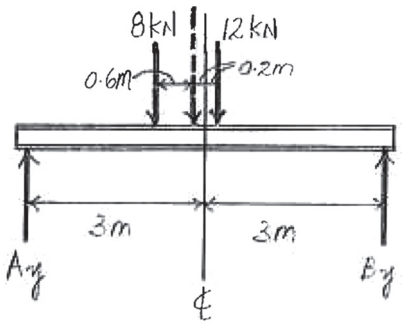
$\circlearrowleft +\Sigma M_A = 0; \quad B_y(6) - 8(2.2) - 12(3.2) = 0 \quad B_y = 9.333\ \text{kN}$

Referring to Fig. $d$,

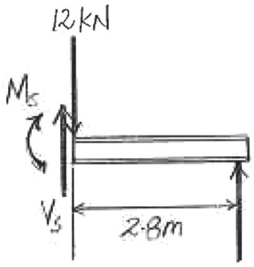
$\circlearrowleft +\Sigma M_S = 0; \quad 9.333(2.8) - M_S = 0 \quad M_S = 26.13\ \text{kN}\cdot\text{m} = 26.1\ \text{kN}\cdot\text{m}\ \text{(Abs. Max.)} \qquad \textbf{Ans.}$




(b)




(c)




(d) $B_y = 9.333\ \text{kN}$

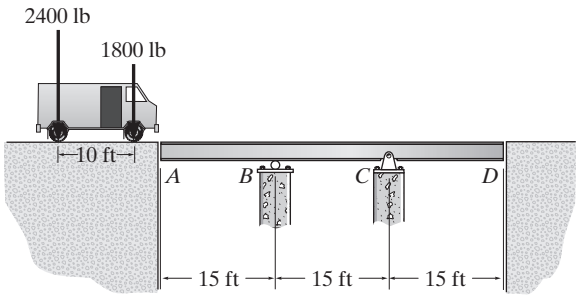
**Ans.**

$V_{\text{max}}^{\text{abs}} = 18.7\ \text{kN};$

$M_{\text{max}}^{\text{abs}} = 26.1\ \text{kN}\cdot\text{m}$

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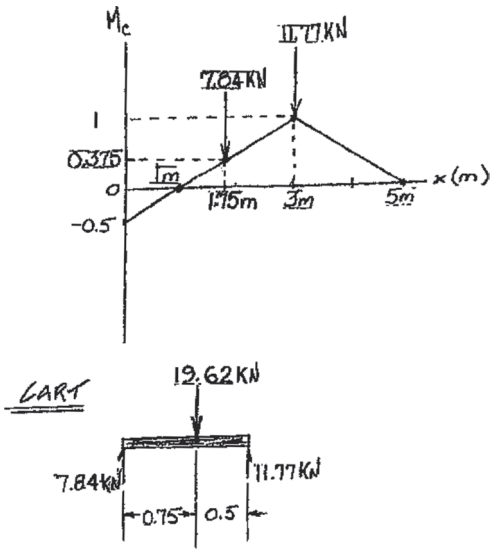
***6–76.** Determine the maximum positive live moment in the girder bridge due to the loading shown. The load is applied directly to the girder.




## SOLUTION

Maximum occurs when wheel $A$ is at $C$.

$$M_{\max} = \frac{1}{2}\left[11.77\ \text{kN}(1) + 7.848\ \text{kN}(0.375)\right] = 7.36\ \text{kN}\cdot\text{m}$$



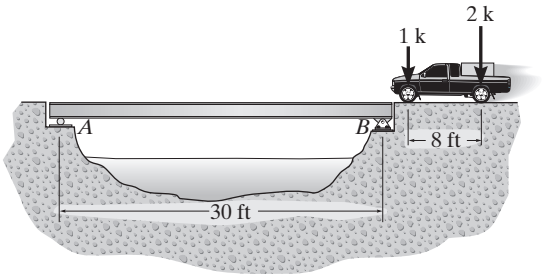




**Ans.**
$M_{\max} = 7.36\ \text{kN}\cdot\text{m}$

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**6–77.** Determine the absolute maximum live moment in the girder bridge due to the loading shown. The load is applied directly to the girder.

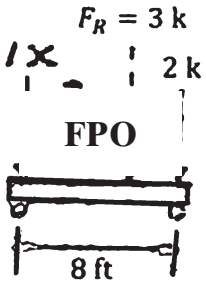


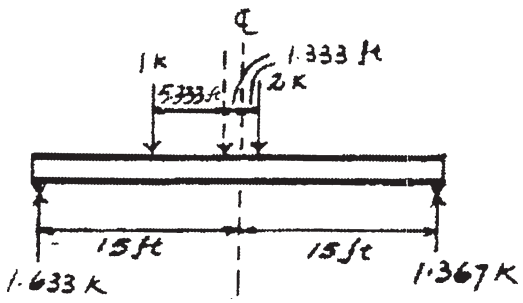

## SOLUTION

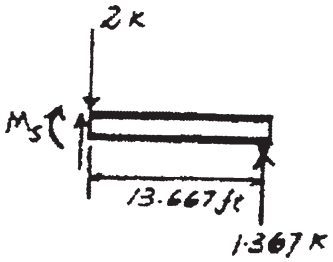
$$\bar{x} = \frac{2(8)}{3} = 5.333 \text{ ft}$$

$$\circlearrowleft + \Sigma M_S = 0; \quad -M_S + 1.367(13.667) = 0$$

$$M_S = 18.7 \text{ k} \cdot \text{ft}$$



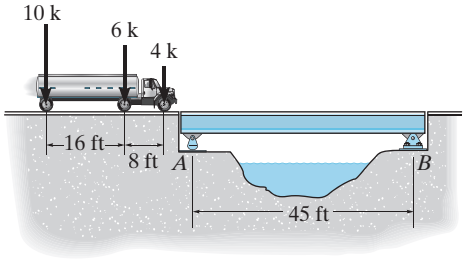



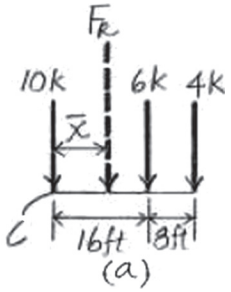

**Ans.**
$M_{\text{abs}}_{\text{max}} = 18.7 \text{ k} \cdot \text{ft}$

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**6–78.** Determine the absolute maximum moment in the beam due to the loading shown.




## SOLUTION




Referring to Fig. *a*, the location of $\boldsymbol{F}_R$ for the moving load is

$+\downarrow F_R = \Sigma F_y; \qquad F_R = 10 + 6 + 4 = 20\ \text{k}$

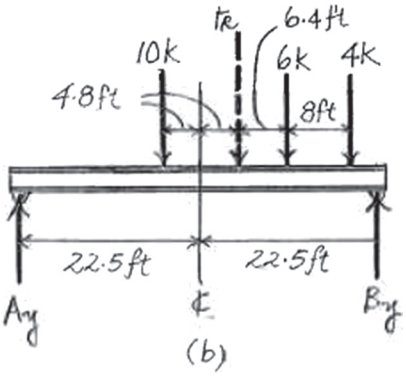
$\circlearrowleft + F_R\bar{x} = \Sigma M_C; \quad -20\bar{x} = -6(16) - 4(24) \quad \bar{x} = 9.60\ \text{ft}$

Assuming that the absolute maximum moment occurs under the 10-k force, Fig. *b*,

$\circlearrowleft + \Sigma M_B = 0; \qquad 4(3.3) + 6(11.3) + 10(27.3) - A_y(45) = 0 \quad A_y = 7.8667\ \text{k}$

Referring to Fig. *c*,

$\circlearrowleft + \Sigma M_S = 0; \qquad M_S - 7.8667(17.7) = 0 \quad M_S = 139.24\ \text{k}\cdot\text{ft} = 139\ \text{k}\cdot\text{ft}$ (Abs. Max.) **Ans.**

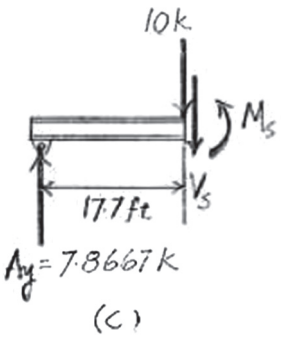


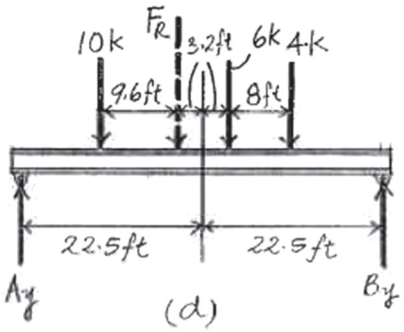

Assuming that the absolute maximum moment occurs under the 6-k force, Fig. *d*,

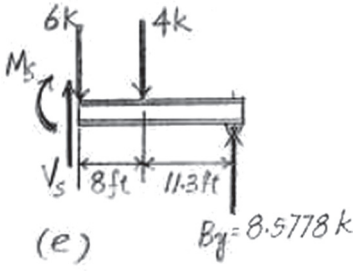
$\circlearrowleft + \Sigma M_A = 0; \qquad B_y(45) - 10(9.7) - 6(25.7) - 4(33.7) = 0 \quad B_y = 8.5778\ \text{k}$

Referring to Fig. *e*,

$\circlearrowleft + \Sigma M_S = 0; \qquad 8.5778(19.3) - 4(8) - M_S = 0 \qquad M_S = 133.55\ \text{k}\cdot\text{ft}$



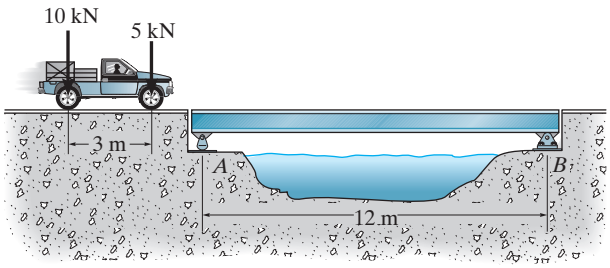




**Ans.**

$M_{\text{abs}}_{\text{max}} = 139\ \text{k}\cdot\text{ft}$

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**6–79.** Determine the absolute maximum shear in the bridge girder due to the loading shown.



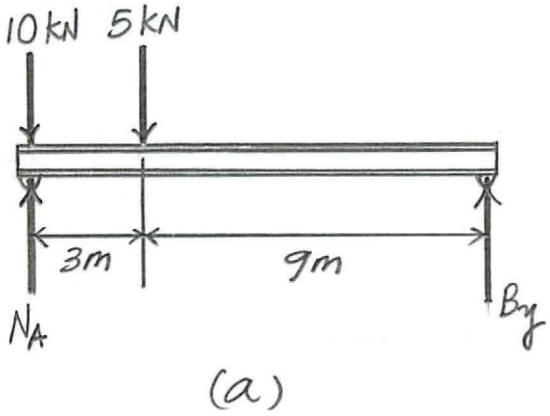

## SOLUTION

By inspection, the absolute maximum shear occurs when the moving loads are in the position with the 10-kN force just to the right of the roller support at $A$, Fig. *a*.

$\circlearrowleft +\Sigma M_B = 0; \quad 5(9) + 10(12) - N_A(12) = 0 \quad N_A = 13.75\ \text{kN}$

Therefore, the absolute maximum shear is

$V_{\text{max}}^{\text{abs}} = N_A = 13.75\ \text{kN}$ **Ans.**

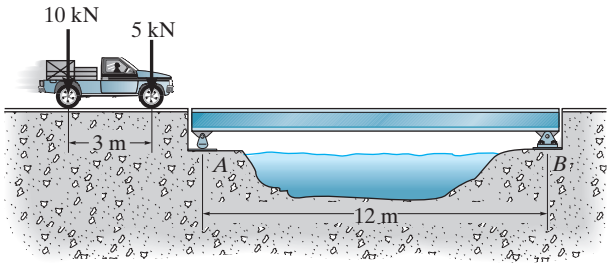



**Ans.**

$V_{\text{max}}^{\text{abs}} = 13.75\ \text{kN}$

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***6–80.** Determine the absolute maximum moment in the bridge girder due to the loading shown.




## SOLUTION

Referring to Fig. *a*, the location of $\boldsymbol{F}_R$ for the moving loads is

$+\uparrow F_R = \Sigma F_y; \quad -F_R = -10 - 5 \quad F_R = 15 \text{ kN}$

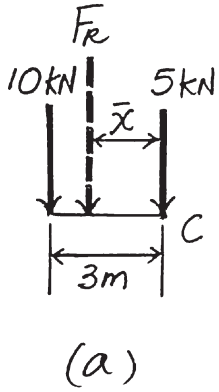
$\circlearrowleft + F_R\bar{x} = \Sigma M_C; \quad 15\bar{x} = 10(3) \quad \bar{x} = 2 \text{ m}$

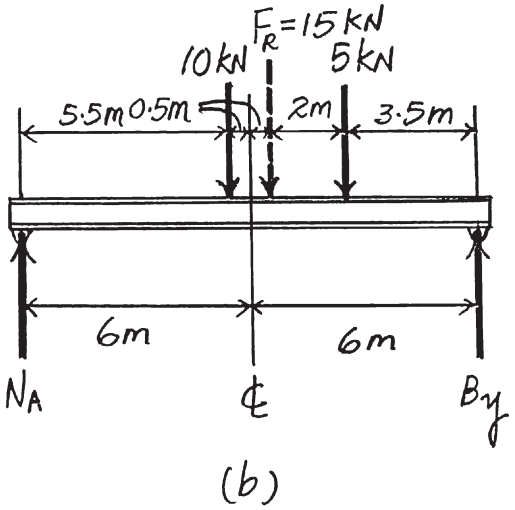
By inspection, the absolute maximum moment occurs under the 10-kN force, Fig. *b*. Write the moment equation of equilibrium about *B*.

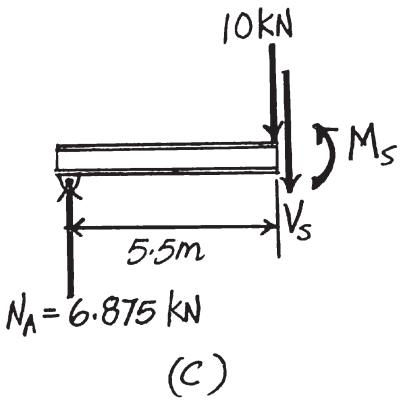
$\circlearrowleft + \Sigma M_B = 0; \quad 15(5.5) - N_A(12) = 0 \quad N_A = 6.875 \text{ kN}$

Referring to the FBD of the left segment of the beam sectioned through $x = 5.5$ m, Fig. *c*,

$\circlearrowleft + \Sigma M_S = 0; \quad M_{\text{abs}}_{\text{max}} = -6.875(5.5) = 0 \quad M_S = M_{\text{abs}}_{\text{max}} = 37.8 \text{ kN} \cdot \text{m}$ **Ans.**



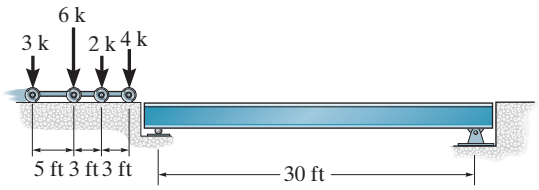




**Ans.**

$M_{\text{abs}}_{\text{max}} = 37.8 \text{ kN} \cdot \text{m}$

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**6–81.** Determine the absolute maximum shear in the beam due to the loading shown.




## SOLUTION

The maximum shear occurs when the moving loads are positioned either with the 3-k force just to the right of the support at $A$, Fig. $a$, or with the 4-k force just to the left of the support at $B$. Referring to Fig. $a$,

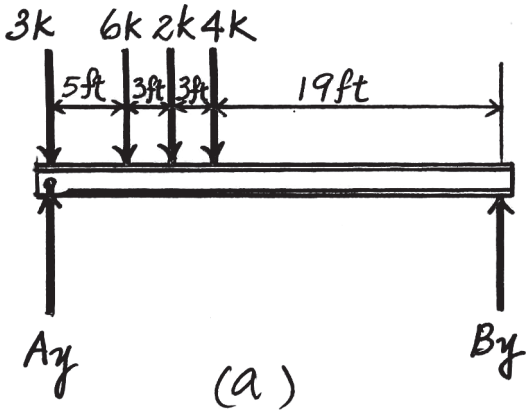
$$\downarrow+\Sigma M_B = 0; \qquad 4(19) + 2(22) + 6(25) + 3(30) - A_x(30) = 0$$
$$A_x = 12.0 \text{ k}$$

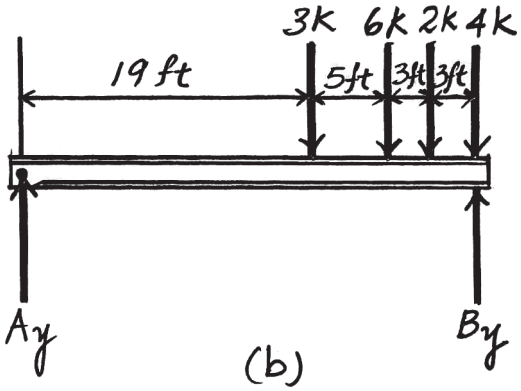
Referring to Fig. $b$,

$$\downarrow+\Sigma M_A = 0; \qquad B_y(30) - 3(19) - 6(24) - 2(27) - 4(30) = 0$$
$$B_y = 12.5 \text{ k}$$

Therefore, the absolute maximum shear occurs for the case in Fig. $b$.

$$V_{\substack{\text{abs}\\\text{max}}} = -B_y = -12.5 \text{ k} \qquad \textbf{Ans.}$$



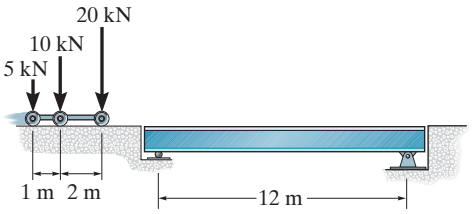




**Ans.**

$$V_{\substack{\text{abs}\\\text{max}}} = -12.5 \text{ k}$$

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**6–82.** Determine the absolute maximum moment in the beam due to the loading shown.




## SOLUTION

Referring to Fig. *a*, the location of $F_R$ for the moving load is

$$+\downarrow F_R = \Sigma F_y; \qquad F_R = 5 + 10 + 20 = 35 \text{ kN}$$

$$\circlearrowleft + F_R\bar{x} = \Sigma M_C; \quad 35\bar{x} = 10(2) + 5(3) \quad \bar{x} = 1.00 \text{ m}$$

Assuming that the absolute maximum moment occurs under the 10-kN force, Fig. *b*,

$$\circlearrowleft + \Sigma M_B = 0; \qquad 20(4.5) + 10(6.5) + 5(7.5) - A_y(12) = 0 \quad A_y = 16.0417 \text{ kN}$$

Referring to Fig. *c*,

$$\circlearrowleft + \Sigma M_S = 0; \qquad M_S + 5(1) - 16.0417(5.5) = 0$$

$$M_S = 83.23 \text{ kN} \cdot \text{m}$$

Assuming that the absolute maximum moment occurs under the 20-kN force, Fig. *d*,

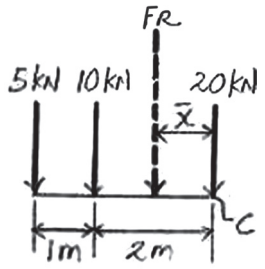
$$\circlearrowleft + \Sigma M_A = 0; \qquad B_y(12) - 5(3.5) - 10(4.5) - 20(6.5) = 0 \quad B_y = 16.0417 \text{ kN}$$

Referring to Fig. *c*,

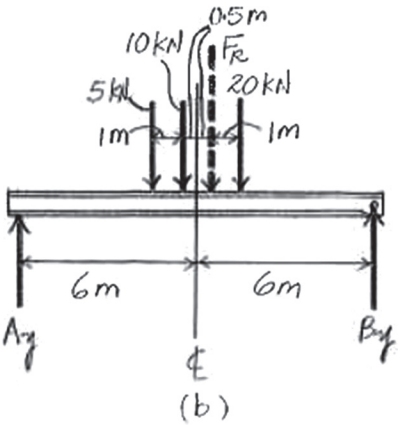
$$\circlearrowleft + \Sigma M_S = 0; \qquad 16.0417(5.5) - M_S = 0$$

$$M_S = 88.23 \text{ kN} \cdot \text{m} = 88.2 \text{ kN} \cdot \text{m} \text{ (Abs. Max.)}$$

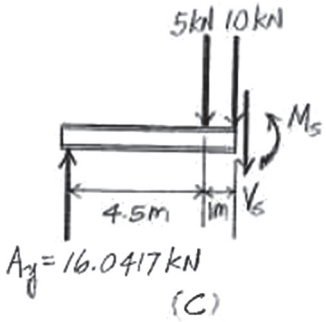
**Ans.**



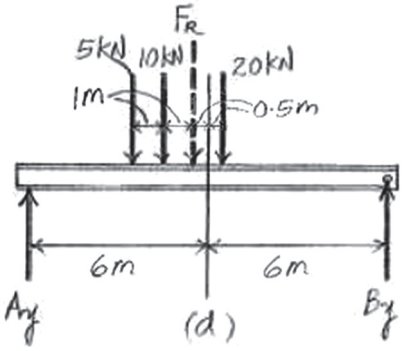

(a)



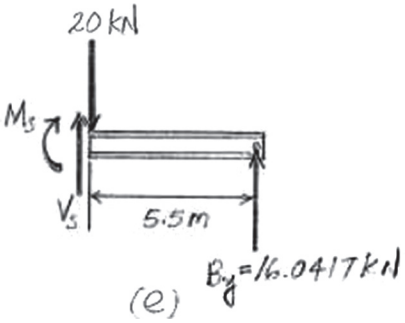

(b)




(c)




(d)



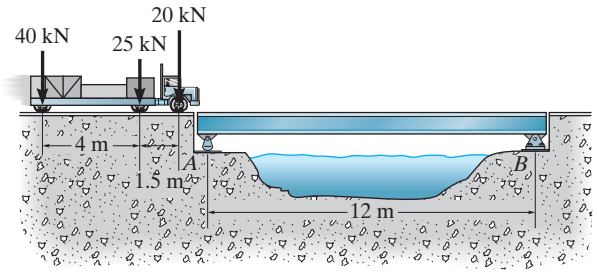

(e)

**Ans.**

$$M_{\substack{\text{abs} \\ \text{max}}} = 88.2 \text{ kN} \cdot \text{m}$$

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**6–83.** Determine the absolute maximum moment in the beam due to the loading shown.



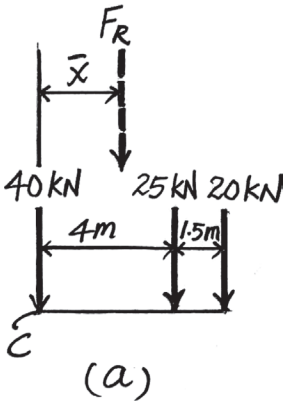

## SOLUTION

Referring to Fig. a, the location of $F_R$ for the moving load is

$$+\downarrow F_R = \Sigma F_y; \quad F_R = 40 + 25 + 20 = 85 \text{ kN}$$

$$\circlearrowleft + F_R\bar{x} = \Sigma M_C; \quad -85\bar{x} = -25(4) - 20(5.5)$$

$$\bar{x} = 2.4706 \text{ m}$$

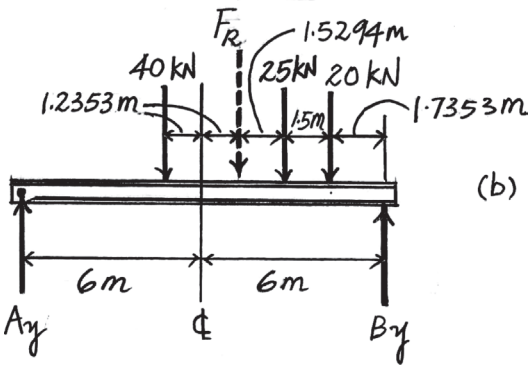


(a)

Assuming that the absolute maximum moment occurs under 40 kN force, Fig. *b*,

$$\circlearrowleft + \Sigma M_B = 0; \quad 20(1.7353) + 25(3.2353) + 40(7.2353) - A_y(12) = 0$$

$$A_y = 33.75 \text{ kN}$$



(b)

Referring to Fig. *c*,

$$\circlearrowleft + \Sigma M_S = 0; \quad M_S - 33.75(4.7647) = 0$$

$$M_S = 160.81 \text{ kN} \cdot \text{m}$$

Assuming that the absolute moment occurs under 25 kN force, Fig. *d*,

$$\circlearrowleft + \Sigma M_A = 0; \quad B_y(12) - 40(2.7647) - 25(6.7647) - 20(8.2647) = 0$$

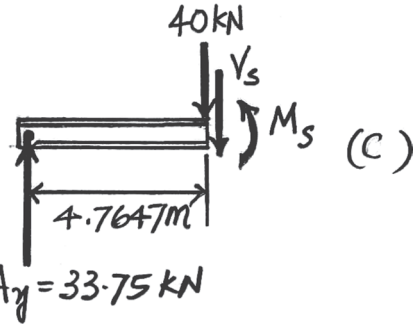
$$B_y = 37.083 \text{ kN}$$

Referring to Fig. *e*,

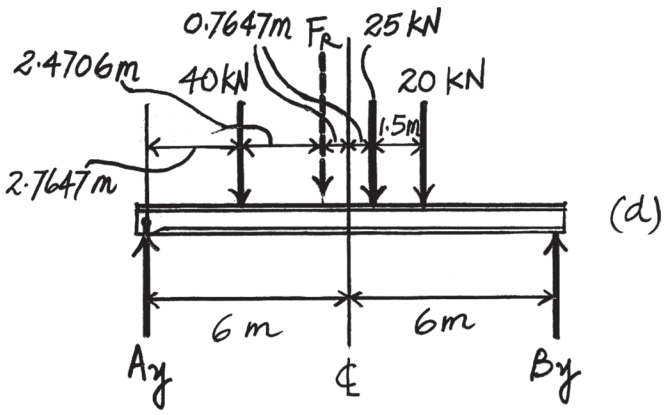
$$\circlearrowleft + \Sigma M_S = 0; \quad 37.083(5.2353) - 20(1.5) - M_S = 0$$

$$M_S = 164.14 \text{ kN} \cdot \text{m} = 164 \text{ kN} \cdot \text{m} \text{ (Abs. Max.)}$$

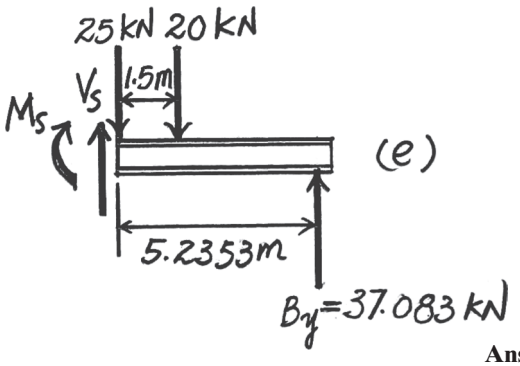
**Ans.**



(c)



(d)



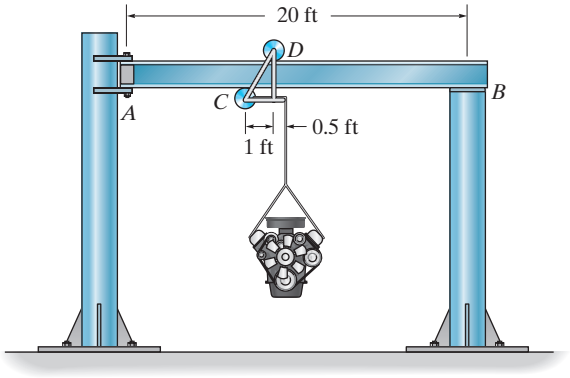
(e)

**Ans.**

$$M_{\underset{\max}{\text{abs}}} = 164 \text{ kN} \cdot \text{m}$$

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***6–84.** The trolley rolls at $C$ and $D$ along the bottom and top flanges of beam $AB$. Determine the absolute maximum moment developed in the beam if the load supported by the trolley is 2 k. Assume the support at $A$ is a pin and at $B$ a roller.




## SOLUTION

Referring to the FBD of the trolley in Fig. *a*,

$\circlearrowleft+\Sigma M_C = 0;$ $\quad N_D(1) - 2(1.5) = 0$ $\quad N_D = 3.00$ k

$\circlearrowleft+\Sigma M_D = 0;$ $\quad N_C(1) - 2(0.5) = 0$ $\quad N_C = 1.00$ k

Referring to Fig. *b*, the location of $F_R$ is

$+\downarrow F_R = \Sigma F_y;$ $\quad F_R = 3.00 - 1.00 = 2.00$ k$\downarrow$

$+\downarrow F_R\bar{x} = \Sigma M_C;$ $\quad -2.00\,(\bar{x}) = -3.00(1)$

$\bar{x} = 1.5$ ft

The absolute maximum moment occurs under the 3.00-k force, Fig. *c*.

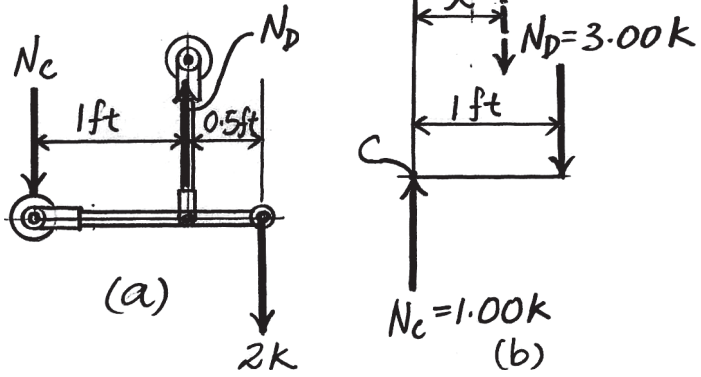
$\circlearrowleft+\Sigma M_A = 0;$ $\quad B_y(20) + 1.00(8.75) - 3.00(9.75) = 0$

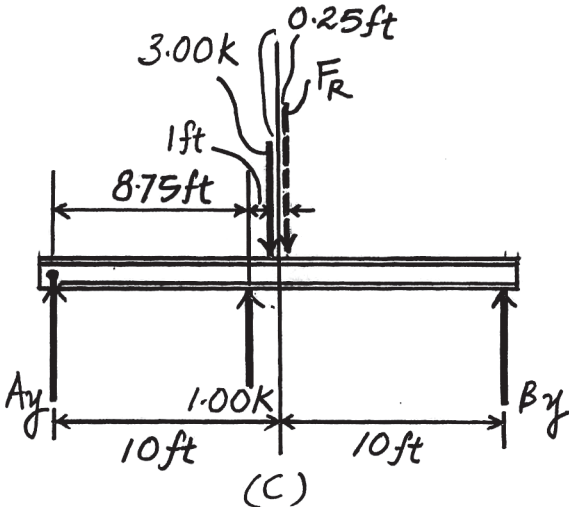
$B_y = 1.025$ k

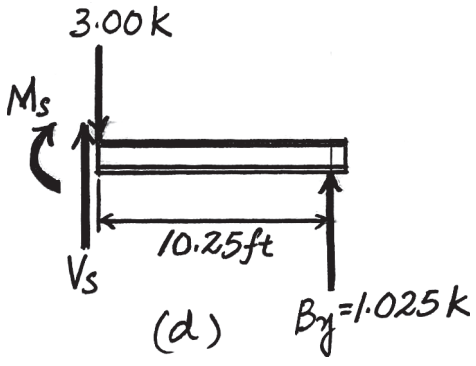
Referring to Fig. *d*,

$\circlearrowleft+\Sigma M_S = 0;$ $\quad 1.025(10.25) - M_S = 0$

$M_S = 10.5$ k · ft (Abs. Max.) **Ans.**



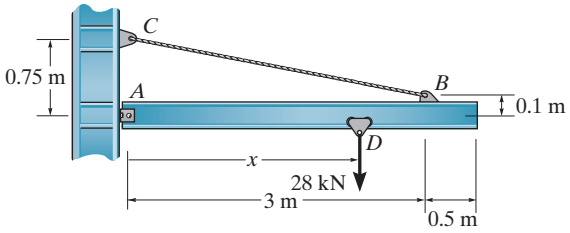



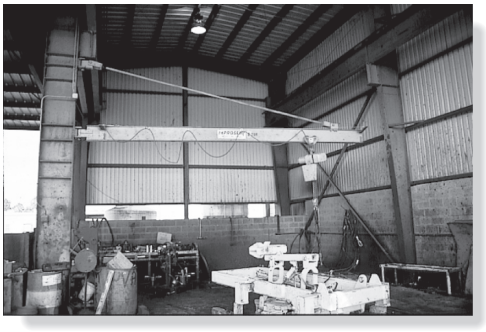




**Ans.**
$M_S = 10.5$ k · ft (Abs. Max.)

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**6–1P.** The chain hoist on the wall crane can be placed anywhere along the boom $(0.1\text{ m} < x < 3.4\text{ m})$ and has a rated capacity of 28 kN. Use an impact factor of 0.3 and determine the absolute maximum bending moment in the boom and the maximum force developed in the tie rod $BC$. The boom is pinned to the wall column at its left end $A$. Neglect the size of the trolley at $D$.



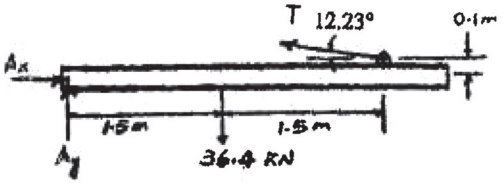
## SOLUTION

Absolute maximum moment occurs when the trolley is at $x = 1.5$ m.

Load $= 28 + 0.3(28) = 36.4$ kN

$\circlearrowleft +\Sigma M_A = 0;\quad T \sin 12.23°(3) + T \cos 12.23°(0.1) - 36.4(1.5) = 0$

$T = 74.49$ kN




$\circlearrowleft +\uparrow \Sigma F_y = 0;\quad A_y - 36.4 + 74.49 \sin 12.23° = 0$

$A_y = 20.63$ kN

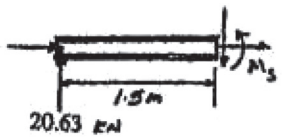
$\circlearrowleft +\Sigma M_S = 0;\quad M_S - 20.63(1.5) = 0$

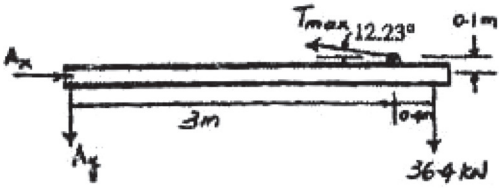
$M_{\max} = 30.9\text{ kN}\cdot\text{m};$ **Ans.**

Absolute maximum tension occurs in the tie rod when trolley $x = 3.4$ m.

$\circlearrowleft +\Sigma M_A = 0;\quad T_{\max} \sin 12.23°(3) + T_{\max} \cos 12.23°(0.1) - 36.4(3.4) = 0$

$T_{\max} = 169$ kN **Ans.**

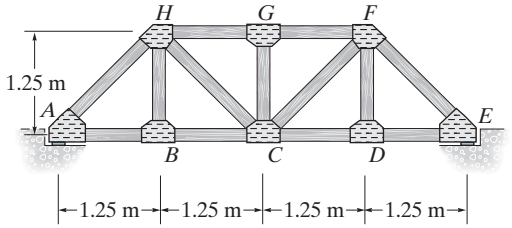


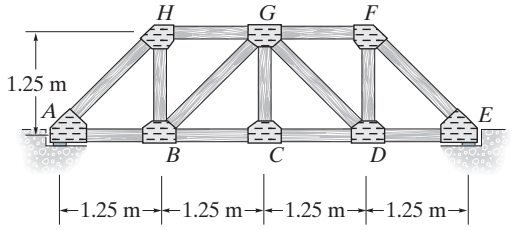

**Ans.**
$M_{\max} = 30.9\text{ kN}\cdot\text{m};$
$T_{\max} = 169$ kN

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**6–2P.** A simply supported pedestrian bridge is to be constructed in a city park and two designs have been proposed as shown in case *a* and case *b*. The truss members are to be made from timber. The deck consists of 1.5-m-long planks that have a mass of 20 kg/m$^2$. A local code states the live load on the deck is required to be 5 kPa with an impact factor of 0.2. Consider the deck to be simply supported on stringers. Floor beams then transmit the load to the bottom joints of the truss. (See Fig. 6–23.) In each case find the member subjected to the largest tension and largest compression load and suggest why you would choose one design over the other. Neglect the weights of the truss members.




case *a*




case *b*

## SOLUTION

Dead load:

$w_d = 20(9.81)(1.5) = 294.3 \text{ N/m}$

Live load:

$w_1 = 5000(1 + 0.2)(1.5) = 9000 \text{ N/m}$

For each truss:

$w_d = \dfrac{294.3}{2} = 147.15 \text{ N/m}$

$w_1 = 4500 \text{ N/m}$

Case a:
Largest compression members are *AH* or *EF*.

$$F_{AH} = F_{EF} = (147.15 + 4500)\left(\frac{1}{2}\right)(1.061)(5) = 12.3 \text{ kN (C)}$$ **Ans.**

Largest tension members are *AB, BC, CD, DE*.

$$F_{AB} = F_{BC} = F_{CD} = F_{DE} = (147.15 + 4500)\left(\frac{1}{2}\right)(0.75)(5) = 8.71 \text{ kN (T)}$$ **Ans.**

Case b:
Largest compression members are *AH* and *EF*.

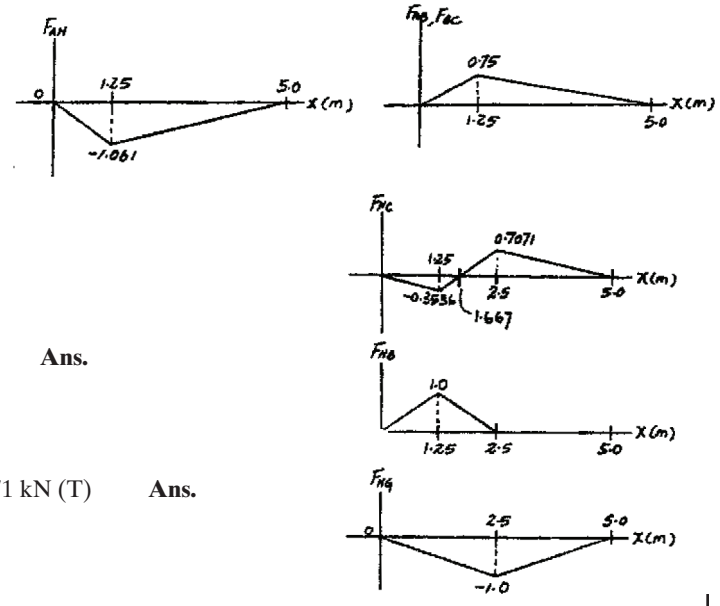
$$F_{AH} = F_{EF} = (147.15 + 4500)\left(\frac{1}{2}\right)(1.061)(5) = 12.3 \text{ kN (C)}$$ **Ans.**

Largest tension members are *BC* and *CD*.

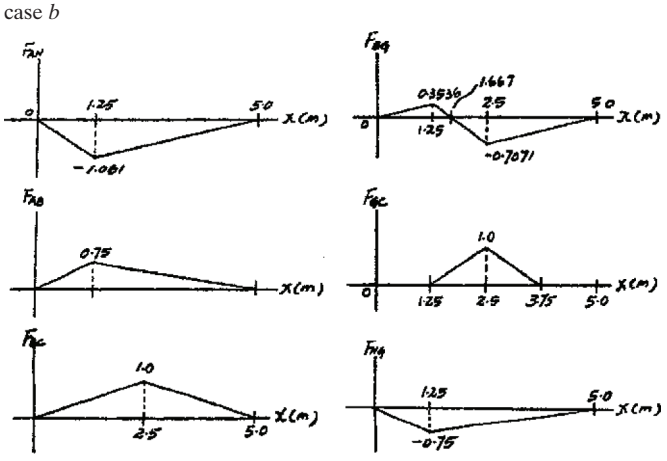
$$F_{BC} = F_{CD} = (147.15 + 4500)\left(\frac{1}{2}\right)(0.1)(5) = 11.6 \text{ kN (T)}$$ **Ans.**

Choose case a to avoid the larger tension forces in member *BC* and *CD*. In timber construction, joints subjected to large tension should be avoided. **Ans.**

case *a*




case *b*




**Ans.**
Case *a*: $F_{\max (C)} = 12.3$ kN; $F_{\max (T)} = 8.71$ kN;
Case *b*: $F_{\max (C)} = 12.3$ kN; $F_{\max (T)} = 11.6$ kN;
Choose case *a*.